

Hours and Wages in Equilibrium*

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Abstract

There are substantial differences in the earnings premium for working long weekly hours across occupations. These are important to understand gender inequality and sorting in the labor market sorting. To explain why these premiums emerge, previous work emphasizes the role of firms' production technologies. But hours and wages are equilibrium outcomes, which also depend on workers' preferences and bargaining power. To study the importance of these forces in shaping long-hour premiums, I build a tractable directed search and matching framework with two-sided multidimensional heterogeneity and endogenous hours and wages. I estimate the model using data on matching patterns, earnings, and vacancies representative for the United States. I find that productivity differences are important in shaping the large earnings premiums in Business occupations. But in Health and Tech occupations, a large share of the long-hour premium is related to bargaining power. Through a counterfactual simulation, I show that removing firms' incentives to reward long hours reduces the gender gap in earnings only marginally. Equalizing workers' valuations of working hours halves the gap.

Keywords: hours, wages, occupations, search and matching

JEL Codes: J31, J32, C78

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1 Introduction

There are substantial differences across occupations in weekly working hours, and in the monthly earnings premium for working long hours. A seminal paper by [Goldin \(2014\)](#) argues that these differences are key to understanding sorting patterns in the labor market, and contribute significantly to gender inequality at the upper end of the earnings distribution. To explain why long-hour premiums vary, [Goldin \(2014\)](#) and [Goldin and Katz \(2016\)](#) emphasize heterogeneity in how hours enter the firm's production function. In some occupations, working long weekly hours generates substantial value for the firm, driven by client expectations or business models that reward continuous employee availability. These productivity differences are then reflected in sizable wage premiums. In other occupations, the productivity premium for firms is substantially lower, which translates into smaller long-hour wage premiums.

Yet, as noted in [Olivetti et al. \(2024\)](#), there is little empirical evidence that pins down firms' production technologies as the primary determinant of the variance in long-hour premiums across occupations. Because hours and wages are equilibrium outcomes, they are shaped not only by productivity differences, but also by workers' preferences and the labor market structure. When long weekly hours are strongly disliked, or when markets are tighter and workers can exploit their favorable bargaining positions, equilibrium long-hour premiums can adjust. We know little about the importance of these different factors. But the effectiveness of policies aimed at reducing long-hour premiums, advocated by [Goldin \(2014\)](#), hinges on the balance between these forces.

This paper introduces a tractable search and matching framework in which hours and wages are jointly determined at equilibrium to quantify how long-hour premiums are shaped, disentangling the contributions of firms' production technologies, workers' preferences, and bargaining power. I consider a pairwise directed search environment similar to [Moen \(1997\)](#), but allow for two-sided multidimensional heterogeneity as in [Choo and Siow \(2006\)](#). Firms post vacancies that specify the wages they are willing to pay for different workers with different characteristics at various hours arrangements. Their postings reflect the productive value of a worker, but also account for the probability of filling the posting. Workers observe all offers, and simultaneously apply to their preferred positions. They consider their preferences over wages, hours, and ameni-

ties, and also account for the probability of getting hired. The interaction of these forces shapes equilibrium hours and wages, and heterogeneous long-hour premiums arise endogenously.

To estimate the model, I build on the approach of [Choo and Siow \(2006\)](#). They introduce an empirically tractable framework to identify frictionless equilibrium models with Transferable Utility (TU), which allows for multidimensional heterogeneity on both sides of the market. I extend their model by introducing search frictions, and allowing for agents to direct their search efforts à la [Moen \(1997\)](#). This addresses two limitations of the baseline model. First, frictions rationalize the coexistence of unemployed workers and vacant jobs as the consequence of coordination failures, rather than voluntary outcomes. Second, in directed search models, agents commit ex ante to searching for a partner with particular characteristics, which defines different submarkets. Because the terms of trade are specific to these submarkets, the matching problem has an Imperfectly Transferable Utility (ITU) structure, as in [Legros and Newman \(2007\)](#) and [Galichon et al. \(2019\)](#). This has important implications, because relaxing the TU assumption means that hours and long-hour premiums are no longer restricted to their efficient surplus-maximizing levels, and instead respond to changes in bargaining power.

Before estimating the model, I document empirically that market tightness affects hours and long-hour premiums. To do so, I combine the latest [American Community Survey \(ACS\)](#) with individual job vacancy postings provided by the [National Labor Exchange \(NLx\)](#) Research Hub. The ACS contains information on wages, matching patterns, and unemployment, while the NLx allows me to construct the occupational distribution of vacant jobs, which differs notably from that of filled jobs. Combining both datasets allows me to measure market tightness for workers and firms of different characteristics across regions. In the high-end labor market of [Goldin \(2014\)](#), I find that a 1% increase in the vacancy-to-unemployment ratio increases wages by up to 0.023 to 0.038%, and reduces hours by 0.012 to 0.020%. But the extent to which wages increase depends on hours worked, and individuals who work longer experience disproportionate wage gains when markets tighten. A 1% increase in the vacancy-to-unemployment ratio increases the long-hour premium by about 0.007%.

I then estimate the model to disentangle how workers' preferences, firms' production technologies, and bargaining power contribute to shaping the equilibrium long-hour premiums. First, across all occupations, a large fraction of the long-hour premium simply arises as a compensa-

tion for workers who prefer working regular instead of long hours. This amenity component explains the entire wage premium in occupations where the observed wage premiums are small, such as the lower earning Health and Science occupations. Second, productivity differences are the next most important determinant of the observed wage premiums. Particularly, in line with the narrative of [Goldin \(2014\)](#), productive gains from long weekly hours explain up to 40% of the monthly earnings premium in high earning Business occupations, such as Lawyers, Judges, Executives, and General Managers. Finally, while differences in market tightness are the least important factor, in some occupations they explain a larger part of the earnings than productive differences. For example, in occupations with high social value, such as Health and Science, the relative excess supply of long weekly hours decreases the long-hour premium by 10 to 20%, while excess demand for long weekly hours increases the earnings premiums of Computer Scientists by almost 10%.

Next, I use the model to assess how the wage structure components, in addition to differential sorting across occupations and weekly hours, contribute to the gender gap in earnings. Sorting into different occupations and hours respectively explain about 35 and 10%, which is close to the estimates of [Blau and Kahn \(2017\)](#) and [Olivetti et al. \(2024\)](#). The model's wage functions reveals two additional mechanisms. First, women sort into jobs with better amenities, which explains about 35% of the earnings gap. This lies in between recent estimates by [Maestas et al. \(2023\)](#) and [Bell \(2024\)](#). Second, women on average sort into tighter markets, for example in Health occupations, which means that they on average have higher bargaining power than men. This compresses the earnings, which would be 20% higher absent this structural advantage. The remaining 37% earnings difference that is not captured by the model stems from differences in (perceived) productivity and other factors.

Finally, I evaluate two policy counterfactuals aimed at increasing gender equality in earnings. The first reduces the firm-side productivity premium attached to long weekly hours. I find that this is surprisingly ineffective, since even eliminating the productivity premium entirely reduces the gender earnings gap by only 5%. While gendered re-sorting across occupations and hours reduces the gap, this is partly offset by women sorting into regular-hour positions with better amenities. The effectiveness of this policy thus increases in tighter markets, where firms cannot exploit their bargaining positions to extract large compensating differentials for amenities.

The second policy equates men and women’s valuations over working hours. This intervention is substantially more effective, and equating their preferences entirely completely more than halves the gender gap in earnings.

These findings have clear policy implications. First, they highlight the importance of taking the market structure into account when designing policy interventions. Second, policies that narrow gender differences in the amenity cost of long hours, such as paid parental leave or affordable high-quality childcare, can be effective. But only insofar as these policies induce genuine convergence in valuations. Recent evaluations of leave- and care-related policies shows mixed evidence, in part because men often take leave alongside women or do not differentially adjust time use ([Andresen and Nix, 2025](#)). This underscores the role of broader shifts in workplace practices and social norms ([Olivetti et al., 2024](#)).

Outline. The remainder of this paper is structured as follows. Section 2 discusses the literature that studies the relation between hours and wages, and compares my framework to related models of matching with and without search frictions. Section 3 introduces the data and presents the empirical patterns that relate hours, wages, and market tightness. Section 4 introduces a tractable equilibrium search and matching model to further interpret the empirical patterns. Section 5 discusses identification and structural estimation of the model. Section 6 presents the estimated parameters and studies how equilibrium hours and wages are shaped. Section 7 concludes.

2 Related Literature

Hours, Earnings, and Gender. This paper contributes to the literature that studies how the relation between hours and earnings contributes to gender inequality in the labor market. A substantial body of work highlights the importance of large wage penalties associated with part-time work ([Macpherson and Hirsch, 1995](#); [Aaronson and French, 2004](#); [Manning and Petrongolo, 2008](#)). More recently, several papers find similar patterns at the opposite of end the distribution, documenting large earnings premiums to long weekly hours ([Bertrand et al., 2010](#); [Goldin,](#)

2014; Cortés and Pan, 2019; Mantovani, 2023).¹ Seminal work by Goldin (2014) and Goldin and Katz (2016) argues that these premiums differ across occupations, reflecting the shape of the production technologies in hours worked. They argue that, because of differential household responsibilities, men disproportionately sort into high-paying and long-hour occupations.

Recent papers by Erosa et al. (2022), Jang and Yum (2022), and Erosa et al. (2024) have embedded nonlinear wage structures into joint models of labor supply and occupation choice. They find that long-hour premiums are important in shaping labor supply decisions and can explain a considerable part of the gender gap in earnings. I differ from these studies in several ways. First, in addition to studying the consequences of workers choosing between occupation with different earnings structures, I explore why varying earnings premiums arise, allowing for multiple determinants. Second, I include rich multidimensional heterogeneity on both sides of the market in an equilibrium model with endogenous hours and wages. As both hours and wages are endogenous, I can study how changes in different primitives affect the equilibrium wage structure, instead of introducing exogenous changes.

Search Models with Amenities. This paper also adds to the literature that studies amenity preferences within search models of the labor market. As noted in Goldin (2014), long-hour premiums can be interpreted as compensating differentials à la Rosen (1986) that reward workers for the disamenity of long weekly hours. But when search frictions are introduced into Rosen (1986)'s model, Hwang et al. (1998) and Lang and Majumdar (2004) show that wages need not reflect the underlying preferences of workers or firms.² I also develop a hedonic framework with search frictions and amenities, but allow agents to direct their search efforts as in Peters (1984), Moen (1997), Eeckhout and Kircher (2010), or Eeckhout and Sepahsalari (2024). Unlike in random search models, wage offers are tied to posted contracts and application choices, which are functions of the model primitives. This implies that equilibrium wages remain linked to preferences and production technologies as in the Rosen (1986) model. But unlike in the competitive equilibrium,

¹Recent work has highlighted that the aggregate relation between hours and wages is better described by a hump shape, with hourly wages decreasing again upon exceeding an occupation-specific threshold (Yurdagul, 2017; Bick et al., 2022).

²When random matching opportunities and rent splitting pin down wages, firms can have opportunities to extract positive profits from offering higher wages and better amenities by reducing turnover. A recent paper by Lavetti and Schmutte (2025) shows that allowing for differentiated firms provides a way to still identify preferences within the random search model of Hwang et al. (1998).

the compensating differential is not equivalent to the marginal worker's valuation, and instead reflects a weighted average of workers' preferences and firms' productivity. The bargaining weight that governs the surplus split is pinned down by tightness in the labor market.

Because of the identification challenges highlighted in [Hwang et al. \(1998\)](#), few empirical applications recover preferences for amenities or working hours within search models. The exceptions typically exploit information on job durations. For example, [Bonhomme and Jolivet \(2009\)](#) find that workers have strong preferences over working times, which translate into moderate compensating differentials because of search frictions. Similarly, [Liu \(2016\)](#) estimates preferences for part-time work in a dynamic search model, focusing on changes in hours worked within job transitions. [Flabbi and Moro \(2012\)](#) instead use a discontinuity in the empirical wage distribution to study preferences for part-time work in a search model with bargaining. They document a tight connection between preferences and labor market outcomes, and find that many women positively value part-time hours, which are costly to provide for firms. Relative to these papers, I estimate an equilibrium model with multidimensional heterogeneity on both sides of the market, and focus on long hours instead of part-time work.

Several recent papers use search models to study the share of wage variation that is attributable to amenities ([Sullivan and To, 2014](#); [Sorkin, 2018](#); [Taber and Vejlín, 2020](#); [Lamadon et al., 2024](#)). These approaches also rely on job transitions and revealed preference arguments. This literature highlights the aggregate importance of compensating differentials, which typically explain between 15 and 30% of earnings inequality. Recent work by [Morchio and Moser \(2024\)](#) uses a similar approach to show that amenities explain up to half of the gender gap in earnings. An important benefit of these approaches is that they require no direct information on amenities. But this comes at the cost of not being able to identify preferences or prices for specific amenities. In contrast, I estimate preferences for hours on both sides of the market, and study how these translate into equilibrium wage premiums. I also provide an alternative approach to identification that uses cross-sectional variation in matching patterns and wages across several markets.

Separable Matching Models. To estimate the model, I build on the seminal work of [Choo and Siow \(2006\)](#) that embeds [McFadden \(1978\)](#)'s random utility model into a two sided matching

framework. They introduce a separability restrictions that allows for the identification of an equilibrium model with two-sided multidimensional heterogeneity. This approach has become the benchmark to empirically study marital sorting.³ But so far, labor market applications remain limited, with exceptions that mainly focus on low friction markets (Dupuy et al., 2020; Dupuy and Galichon, 2022; Corblet, 2023; Dupuy et al., 2023). An important reason is that the model assumes away search frictions, which is at odds with the large literature that followed Diamond (1982), Mortensen (1982), and Pissarides (1985).

In adding search frictions into a separable matching model, this paper is related to Jaffe and Weber (2019), Ciscato (2024), and San (2024). They introduce heterogeneous meeting costs between agents of different types into models based on Choo and Siow (2006). An important difference is that I assume agents search and match in local markets, as in standard directed search models (Moen, 1997; Chade et al., 2017). Since agents fully commit to their chosen local market, this imposes no centralized market clearing condition. This has nontrivial implications in terms of utility being imperfectly transferable (ITU) between workers and firms (Legros and Newman, 2007; Galichon et al., 2019). By relaxing the usual (perfectly) transferable utility (TU) restriction, I allow for market tightness to affect endogenous matching terms beyond wages. Similar extensions have recently been introduced in Mourifié (2019) and Mourifié and Siow (2021), but without a clear structural underpinning.

Related work by Arcidiacono et al. (2016) and Beauchamp et al. (2024) also considers directed search models in a separable matching model based on Choo and Siow (2006). But they study the marriage market, where information on transfers is not readily available. To identify preferences on both sides of the market, they assume that utility cannot be transferred between potential partners (NTU). While this can be justified in particular settings, it is a strong assumption in modern labor (marriage) markets, because it rules out any concession between workers and firms (men and women). By relaxing this assumption and allowing for utility to be imperfectly transferable, this paper widens the potential scope of applications.

³Some notable recent examples of studies that use (extensions of) their approach to study marriage markets are Dupuy and Galichon (2014), Choo (2015), Siow (2015), Greenwood et al. (2016), Chiappori et al. (2017), Chiappori et al. (2018), Ciscato and Weber (2020), Chiappori et al. (2020), Galichon and Salanié (2022), Ciscato (2024), Reynoso (2024), and Chiappori et al. (2025).

3 Empirical Patterns

This section introduces the data used to analyze the relation between hours, earnings, and tightness in the labor market. Individual workers and occupations are first classified into discrete types based on their observable characteristics. I show that hours and earnings differ substantially across these types, and study heterogeneity in the returns to long hours across occupations as in [Goldin \(2014\)](#) and [Cortés and Pan \(2019\)](#). Next, I highlight differences in the occupational distributions of filled and vacant positions, and estimate how hours, earnings, and long-hour premiums respond to changes in the vacancy-to-unemployment ratio.

Data. The primary data source is the latest wave of the [American Community Survey](#) (ACS, 2023), provided by the [Integrated Public Use Microdata Series](#) (IPUMS). I use the ACS to construct a sample of the high-end labor force in the United States as in [Goldin \(2014\)](#). The sample consists of individuals between 25 and 64 years old who have at least a bachelor’s degree and are, or were, employed in a high earning occupation as listed in Table 1. I construct an employed sub-sample of only full-time workers.⁴ Additionally, I also construct an unemployed sub-sample using individuals who are currently unemployed but who used to work in a high-earning occupation. This assures that the type distribution is representative for the high-end labor market.

The second main data source consists of individual job vacancies, and is provided by the [National Labor Exchange](#) (NLx) Research Hub. They collect online postings from companies’ career websites, state job banks, and the federal jobs portal. The NLx attempts to cover all online vacancies in the United States, and represents roughly half of all openings as estimated in the [Job Openings and Labor Turnover Survey](#) (JOLTS) by the [Bureau of Labor Statistics](#) (BLS). Comparable commercial vacancy data with similar coverage has recently been used by [Deming and Noray \(2020\)](#) and [Acemoglu et al. \(2022\)](#). I use the NLx data to construct the occupational vacancy distributions across regions, and scale these using adjusted JOLTS estimates from [Mongey and Horwich \(2023\)](#). Appendix B.3 provides further information.

⁴Working full-time is defined as at least 35 hours per week, as in [Cortés and Pan \(2019\)](#). Because of the focus on college educated workers in high-earning occupations, there is little part-time employment. Approximately 11% of individuals reports working less than 35 hours per week or 49 weeks per year.

Table 1: Types of Occupations

Business	Health	Tech	Science	Other
Lawyers and Judges Exec's and GMs Managers Financial Ops Business Ops Sales	Medical Practitioners Adv. Clin. Prof. Clin. Prof.	Comp. Programmers Engineers Cybersec. & Infrastr.	Natural Scientists Social Scientists	Training & Teaching Other

Notes. Classification of occupations into types similar to those introduced in [Goldin \(2014\)](#). The individual occupations within each type can be found in Appendix B.1.

Constructing Types. Workers, firms, and hours are grouped into discrete categories (= types) based on observable characteristics. I define $X = 8$ worker types, which classify individuals according to their gender, age, and education level. The age dimension separates workers that are older or younger than forty, and the education level makes a distinction between those with a Bachelor's (BA), and those with at least a Master's degree (MA⁺). I also define $Y = 16$ occupation types, following the broad classification of [Goldin \(2014\)](#) that distinguishes Business, Health, Tech, Science, and Other occupations. The different occupation types are shown in Table 1, and the detailed individual occupations within each type can be found in Appendix B.1. Finally, I also define $Z = 2$ hours types, which classify workers based on whether they work regular (35 to 49) or long (50 or more) weekly hours, defined as in [Cortés and Pan \(2019\)](#). Importantly, average hours within the regular- and long-hour types are comparable across occupations, at about 40 and 55 hours per week respectively, as shown in Appendix B.2. There are thus $X \times Y \times Z = 256$ type combinations, observed across $S = 9$ independent regions.⁵

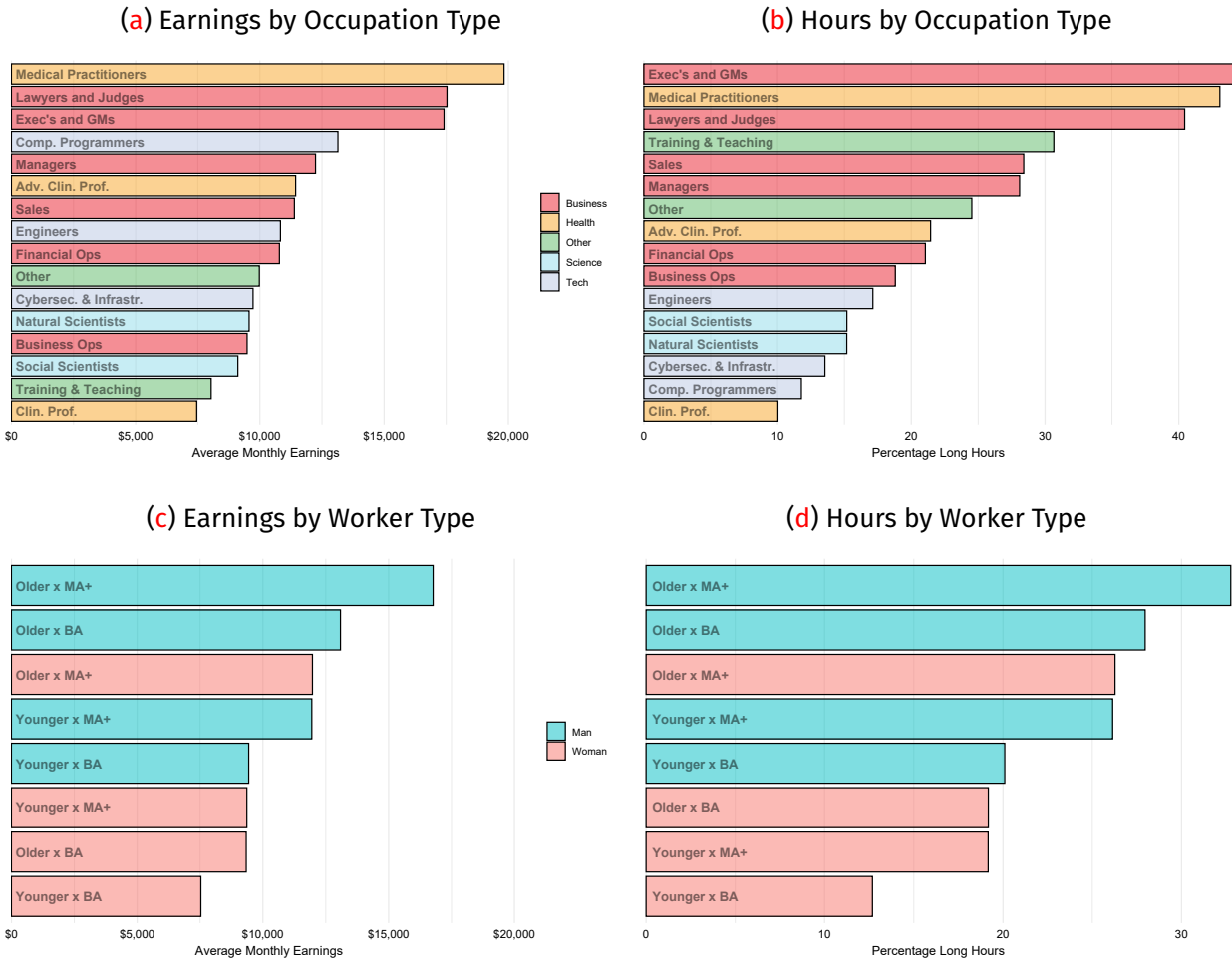
3.1 Hours and Earnings

Occupation Types. Figure 1 shows the variation in hours and monthly earnings across the different worker and occupation types. Panel (a) highlights earnings differences across occupations. The highest earning groups in the sample are Medical Practitioners, Executives, Lawyers, and Judges, who all earn between 17,000 and 20,000 dollars per month. Just below them are upper-level Business occupations, such as Managers, Sales, and Financial Operations, as well as Computer Programmers, Engineers, and Advanced Clinical Professionals, who earn between 10,000

⁵States are aggregated into Census Divisions as defined by the [Census Bureau](#) to be consistent with the model and have sufficient observations per type.

and 13.500 dollars per month. At the lower end, Clinical Professionals, and Training & Teaching occupations have average monthly salaries of around 7.500 dollars. The large wage differences across occupations have been shown to account for an increasing fraction of the gender gap in earnings (Blau and Kahn, 2017; Olivetti et al., 2024).

Figure 1: Earnings and Hours by Type



Notes. Earnings and hours by worker and occupation type in the United States. Data from the American Community Survey (ACS, wave 2023). Panels (a) and (c) show the distribution of monthly earnings across the occupation and worker types defined in section 3. Panels (b) and (d) show the fraction of workers that work long hours, defined as more than 50 per week, across the same types.

Panel (b) highlights differences in hours across the same occupation types. Working long hours is particularly common in the highest earning occupations, being Executives, Lawyers, Judges, and Medical Practitioners. Among them, 40 to 45 percent reports working more than 50 hours per week. In lower earnings occupations, such as Clinical Professionals, Social, and

Natural Scientists, working long hours is relatively unusual. Among them, 15 percent or less reports working long weekly hours. While higher earning occupations are generally characterized by longer hours, there are important exceptions. For example, in Tech occupations only a small fraction of individuals reports long hours, yet their monthly earnings are among the highest. Training and Teaching occupations show the opposite pattern.

Worker Types. Panel (c) shows earnings differences across worker types. This highlights that older men are the highest paid group in the sample, irrespective of education levels. Their average monthly earnings range from 12,500 (BA) to almost 17,500 dollars (MA⁺). Older women and younger men with MA⁺ education follow, with both groups earning close to 12,000 dollars per month. The remaining groups earn less, averaging between 7,500 and 9,000 dollars per month. These earnings differences by gender and education are well known, see for example the review by [Blau and Kahn \(2017\)](#).

Panel (d) highlights differences in working hours across worker types. The first main pattern is that, within education and experience levels, men are more likely to work long hours than women. This difference is particularly pronounced for older workers, among whom almost 30 percent of men with BA education works long hours, increasing to almost 35 percent among MA⁺ educated men. The corresponding percentages among women of the same types are closer to 20 and 25 percent respectively. The second main pattern is that younger women are very unlikely to work long hours. Particularly, if they obtained only a BA degree, less than 15 percent reports working long hours. These patterns are in line with the gender gap in hours among full-time workers presented in [Cortés and Pan \(2019\)](#).

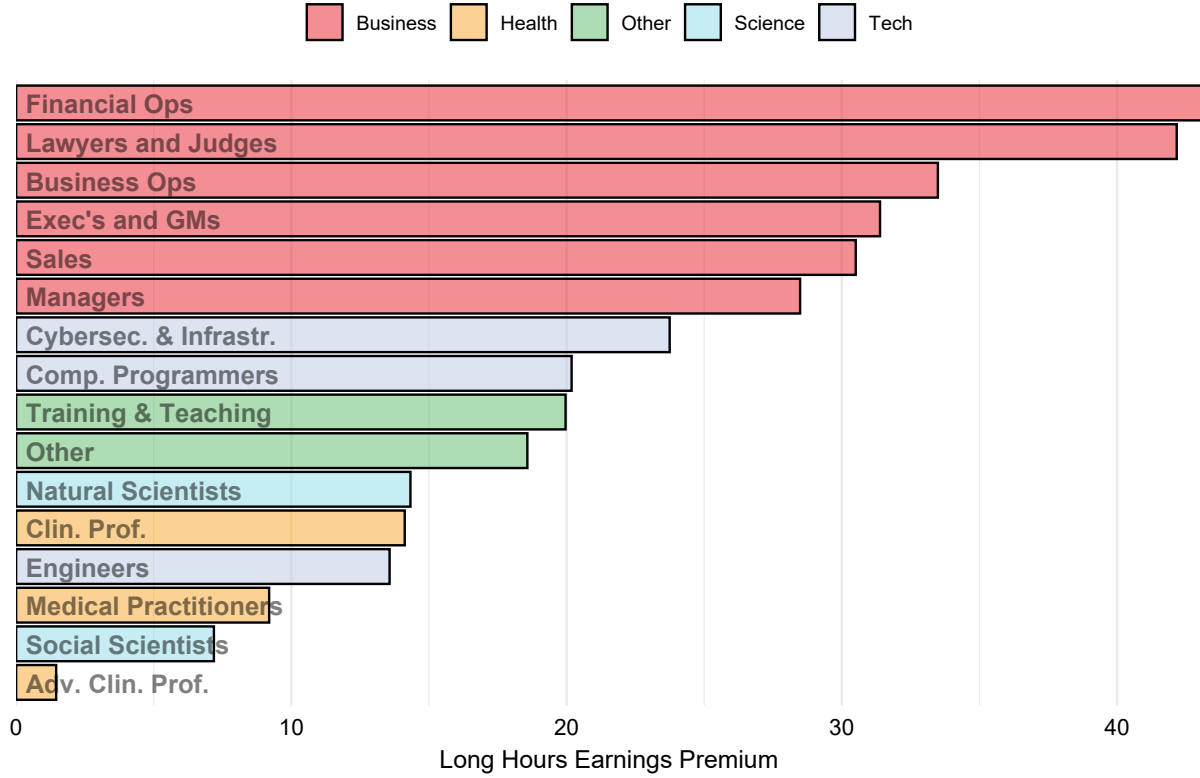
3.2 Long-Hour Premiums

To better understand the equilibrium relation between hours and earnings, I study how the long-hour premium varies across occupations. I do so by estimating regressions similar to those in [Goldin \(2014\)](#) and [Cortés and Pan \(2019\)](#). For each occupation type $y \in Y$, I estimate:

$$\log(W_i) = \alpha + \beta \mathbb{1}_{z_i} + \sum_x \mathbb{1}_{x_i} \gamma_x + \sum_s \mathbb{1}_{s_i} \gamma_s + \varepsilon_i \quad \forall i \in y, \quad (3.1)$$

where W_i is the monthly wage of individual i , $\mathbb{1}_{z_i}$ an indicator for working long hours, $\mathbb{1}_{x_i}$ are worker type fixed effects, and $\mathbb{1}_{s_i}$ region fixed effects.

Figure 2: Returns to Hours by Occupation



Notes. This figure shows the percentage monthly wage premiums for working long hours across occupations, defined as the β coefficients in equation (3.1). The data is from the [American Community Survey](#) (ACS, wave 2023).

Figure 2 shows the long-hour premiums, as captured by the β coefficients from Equation (3.1), across occupations. I find the largest premiums in Business occupations, and particularly among Lawyers, Judges, and in Financial Operations. They earn over 40% more when working long instead of regular weekly hours. The other Business occupations follow, with substantial long-hour premiums varying between 27 and 35%. Tech occupations have the largest earnings premiums outside of Business, varying between 20 and 25% for Computer Programmers and Cybersecurity & Infrastructure occupations respectively. The long-hour premiums are lowest in Health and Science occupations, where working long weekly hours is associated with earning premiums of less than 15%. These patterns are very similar to those reported [Goldin \(2014\)](#) and [Cortés and Pan \(2019\)](#). A notable pattern that emerges is that long-hour premiums are not nec-

essarily informative about occupational earnings. For example, Medical Practitioners, Advanced Clinical Professionals, or Engineers are among the highest earning occupations, but have the lowest premiums for working long hours.

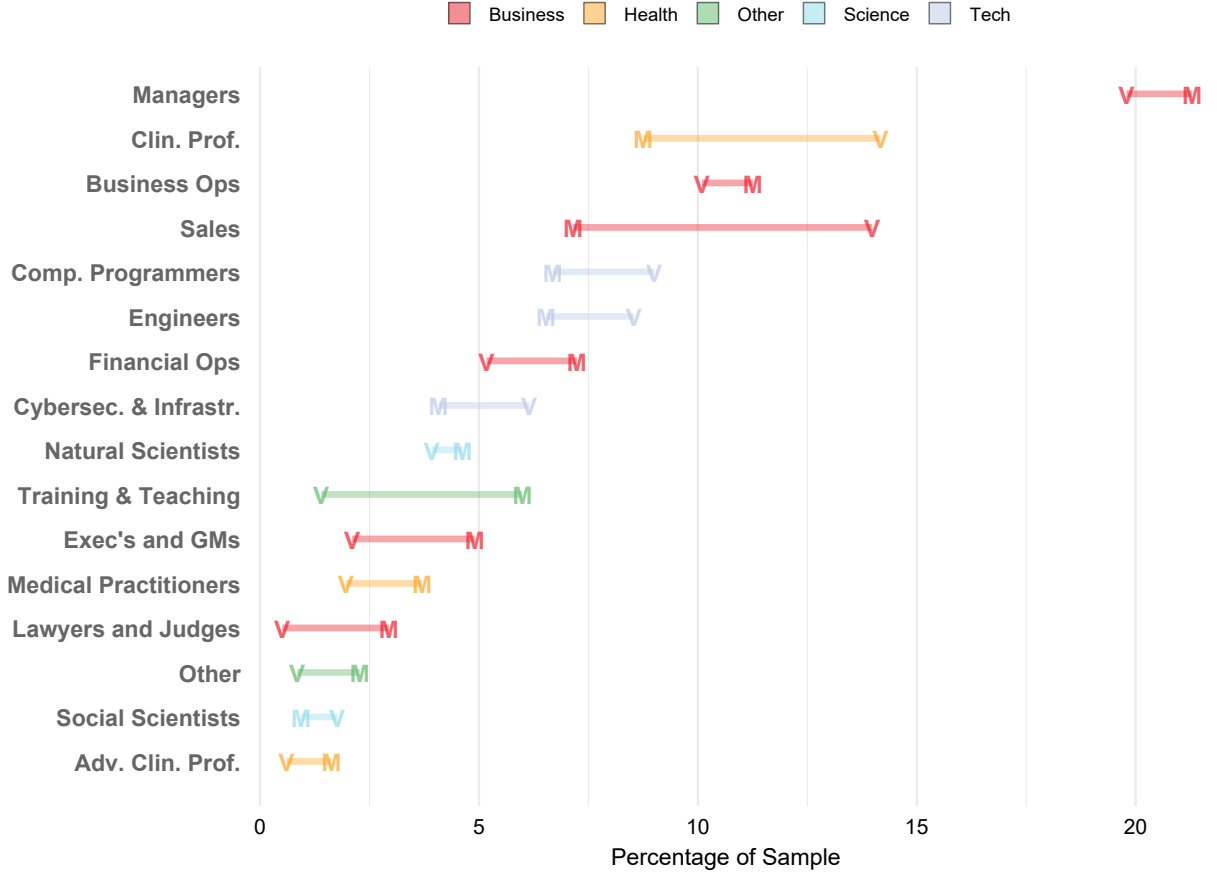
3.3 Filled and Vacant Positions

Figure 3 examines the distributions of filled and vacant jobs across occupations. The first main pattern is the substantial variation in the relative size of different occupations. For example, Business roles such as Managers, Sales, and Business and Financial Operations, account for a large share of both matches and vacancies. They jointly make up over 45% of both filled and vacant positions. In contrast, Social Scientists, Advanced Clinical Professionals, and Other occupations together represent less than 5 percent of observations in either group. The relative sizes of different occupations in *filled* positions aligns well with [Cortés and Pan \(2019\)](#). While there is little comparable evidence on *vacancies*, the distribution resembles that reported in the Appendix of [Hershbein and Kahn \(2018\)](#). Because they study a different period and use an alternative classification of occupations, numbers are nonetheless difficult to compare. The vacancy distribution is also in line with [Bureau of Labor Statistics \(BLS\)](#) projections of occupational growth.⁶

A second notable pattern is the gap between the occupational distributions of vacant and filled jobs. Some occupations make up a substantially larger share of all vacancies than they do of filled positions, suggesting a considerable excess in demand. The main examples are Clinical Professionals, Sales, and all Tech occupations. While these together account for over half of all vacancies, they represent less than a third of matches. In other occupations we see the inverse pattern, as they make up a larger share of filled positions than of vacancies. These include high-earning occupations, such as Financial Operations, Executives, Lawyers, Judges, and Medical Practitioners, and lower earning Training & Teaching occupations, indicating a relative excess in supply. Together, these results highlight the importance of introducing information about the distribution of vacancies, which does not necessarily resemble that of filled jobs.

⁶For example, [Bureau of Labor Statistics \(2024\)](#) includes Nurses, Data scientists, and Information Security Analysts among the 5 fastest growing occupations.

Figure 3: The Distributions of Filled and Vacant Jobs



Notes. This figure highlights the occupational distributions of matched jobs (M) and vacancies (V). Constructed using [National Labor Exchange \(NLx\) Research Hub](#) and [American Community Survey \(ACS\)](#) data.

3.4 Wages, Hours, and Tightness

I now examine how hours and wages relate to labor market tightness, as measured by the occupation- and worker-type specific vacancy-to-unemployment ratio. Specifically, I estimate cross-sectional regressions of the form:

$$\log(Y_i) = \alpha + \tau \log(t_i) + \sum_x \mathbb{1}_{x_i} \gamma_x + \sum_y \mathbb{1}_{y_i} \gamma_y + \sum_s \mathbb{1}_{s_i} \gamma_s + \varepsilon_i, \quad (3.2)$$

where Y_i denotes monthly wages or weekly hours and t_i is the ratio of type y vacancies to unemployed workers of type x within region s .⁷ I also include fixed effects for worker types ($\mathbb{1}_{x_i}$), occupation types ($\mathbb{1}_{y_i}$), and regions ($\mathbb{1}_{s_i}$). The main parameter of interest τ is identified through

⁷Similar results obtain when log hours is replaced with an indicator for working long hours.

within-type variation in market tightness across regions.

To address potential concerns about reverse causality or simultaneity, for example due to local productivity shocks, I construct a leave-one-out instrument for tightness similar to [Azar et al. \(2022\)](#) and [Börschlein et al. \(2024\)](#). The instrument for observations in region s is constructed as:

$$t_i^{-s} = \frac{1}{S-1} \sum_{r \neq s} \log \left(\frac{\hat{M}_{y0}^r}{\hat{M}_{x0}^r} \right), \quad (3.3)$$

where $\hat{M}_{y0}^r$ denotes vacant positions of type y and $\hat{M}_{x0}^r$ are unemployed workers of type x in region r . By excluding the local, within-region conditions, the instrument exploits plausibly exogenous variation in supply and demand at the national level.

Table 2: Wages, Hours, and Tightness

	$\log(W_i)$	$\log(H_i)$	$\log(W_i)$	$\log(H_i)$
	OLS		IV	
$\log(t_i)$	0.023 (0.005)	-0.012 (0.005)	0.038 (0.007)	-0.020 (0.007)
Constant	8.793 (0.011)	1.843 (0.010)	8.813 (0.012)	1.832 (0.012)
Occupation FE	✓	✓	✓	✓
Worker Type FE	✓	✓	✓	✓
Region FE	✓	✓	✓	✓
First Stage F			316,690	316,690
R ²	0.179	0.085	0.179	0.085
Num. obs.	274,544	274,544	274,544	274,544

Notes. This table reports results from estimating equations (3.2) without and with the leave-one-out instrument for market tightness. The data is from the [American Community Survey \(ACS\)](#) and [National Labor Exchange \(NLx\)](#) Research Hub. Bold faced estimates are significant at the 5% level.

Table 2 reports the results from this analysis. The first column shows an estimated elasticity of wages with respect to tightness of 0.023, which increases to 0.038 when instrumenting tightness. The positive sign reflects that workers have higher bargaining power in tighter markets and can negotiate better wage offers. The magnitude of the effect is in line with the calibrated value of 0.030 in [Hagedorn and Manovskii \(2008\)](#). The estimates are also similar to [Hoeck \(2022\)](#) and [Börschlein et al. \(2024\)](#), who find values between 0.010 and 0.020 in Germany and Denmark

using similar estimation strategies. Slightly lower estimates in European countries would be consistent with more centralized collective bargaining and stricter employment protection rules compressing wage adjustments.

The second column of Table 2 shows an opposite elasticity of hours with respect to tightness. The baseline estimate is -0.012, which increases to -0.020 when instrumenting market tightness. The sign is again consistent with workers having more bargaining power in tighter markets, allowing them to negotiate contracts with fewer hours. While there is no direct comparison for these results, recent work by Autor et al. (2023) has highlighted that labor market outcomes beyond wages respond to changes in tightness. But they focus the relation between market tightness and where people work, instead of how long.

Table 3: Tightness and the Long-Hour Premium

	$\log(W_i)$	$\log(W_i)$
	OLS	IV
$\log(t_i)$	0.030 (0.006)	0.026 (0.008)
$\log(H_i)$	0.183 (0.002)	0.190 (0.002)
$\log(t_i) \times \log(H_i)$	-0.002 (0.002)	0.007 (0.002)
Constant	8.457 (0.011)	8.463 (0.013)
Occupation FE	✓	✓
Worker Type FE	✓	✓
Region FE	✓	✓
R ²	0.207	0.207
Num. obs.	274,544	274,544

Notes. Results from estimating equation (3.2) with wages as the outcome, adding an additional hours control and interaction with tightness. The data is from the American Community Survey (ACS) and National Labor Exchange (NLx) Research Hub. Bold faced estimates are significant at the 5% level.

To study the relation between the long-hour premium and market tightness, I extend equation (3.2) by including hours and an interaction between hours and tightness. Table 3 presents

the results from this analysis. The main object of interest is the coefficient on the interaction between tightness and hours, which captures how changes in tightness affect the returns to longer hours. The baseline regression in the first column has an interaction close to zero, which is statistically insignificant. But the effect again increases considerably when we instrument market tightness. The second column shows a significant and positive interaction of about 0.007, with the baseline effect reducing slightly from 0.030 to 0.026. These results highlight that hours, wages, and long-hour premiums all respond significantly to market tightness.

4 Directed Search and Matching with Endogenous Hours

4.1 Utility and Profit Functions

To better understand what shapes long-hour premiums, I introduce a tractable equilibrium search and matching framework with two-sided heterogeneity that endogeneizes hours, wages, and bargaining power. Consider a labor market populated by workers $i \in I$ and firms $j \in J$, who can match one-to-one or remain unassigned.⁸ Workers are grouped into types $x = (1, 2, \dots, X)$, firms into types $y = (1, 2, \dots, Y)$, and hours arrangements into types $z = (1, 2, \dots, Z)$. All types are discrete and can be multidimensional.

Labor Supply. I model the utility of a worker i of type x searching for a type y job with z hours as the sum of a systematic component and an idiosyncratic search cost:

$$u_{x_izy} = p(\theta_{xyz})(\alpha_{xyz} + W_{xyz}) + \xi_{x_izy}. \quad (4.1)$$

The systematic term is the product of a worker's job finding probability $p(\theta_{xyz})$ and their valuation of the job, which is the sum of amenities α_{xyz} and the wage W_{xyz} . Job finding probabilities are a Constant Elasticity of Substitution (CES) function of market tightness as in [Menzio and Shi \(2010\)](#):

$$p(\theta_{xyz}) \equiv (1 + \theta_{xyz}^\lambda)^{\frac{1}{\lambda}}, \quad (4.2)$$

⁸In one-to-one matching models, job and firm are interchangeable. This is equivalent to firms hiring multiple workers if the output is separable across jobs.

where market tightness θ_{xyz} is the ratio of total firms to workers searching in market (x, y, z) , and λ is the elasticity of substitution. Assuming that λ is negative assures that the job finding probability function maps to the unit interval and is concave increasing in tightness. The search costs ξ_{xyz} reflect idiosyncratic differences in application time, effort, or connections across different types of jobs. They rationalize why identical workers sort into different job types and hours arrangements.

Labor Demand. The profit functions of firms include a systematic component and an idiosyncratic screening cost:

$$\pi_{xyz} = q(\theta_{xyz})(\rho_{xyz} - W_{xyz}) + \eta_{xyz}. \quad (4.3)$$

The systematic term is the product of firms' hiring probabilities and workers' productivity net of wages. Hiring probabilities are a decreasing function of market tightness, and are defined by $q(\theta_{xyz}) = \frac{p(\theta_{xyz})}{\theta_{xyz}}$. When hired, workers generates a value ρ_{xyz} for the firm, and are paid a wage W_{xyz} . The screening costs η_{xyz} capture idiosyncratic differences in compatibility or recruitment networks.

4.2 Equilibrium Wages and Matches

As in [Wright et al. \(2021\)](#), I solve for wages by writing the equilibrium problem as firms maximizing profits (4.3) subject to offering workers their expected equilibrium utilities (4.1):

$$\begin{aligned} \max_{W, \theta, x, z} \quad & q(\theta_{xyz})(\rho_{xyz} - W_{xyz}) + \eta_{xyz} \\ \text{s.t.} \quad & u_{xyz} = p(\theta_{xyz})(\alpha_{xyz} + W_{xyz}). \end{aligned} \quad (4.4)$$

By substituting in the constraint and solving for optimal market tightness θ_{xyz}^* , I recover a familiar equilibrium wage function W_{xyz}^* that splits up the total match surplus $\Phi_{xyz} \equiv \alpha_{xyz} + \rho_{xyz}$. The derivation is provided in [Appendix A.1](#). Simplifying notation by omitting the superscripts, the equilibrium wage function writes:

$$W_{xyz} = [1 - \mathcal{F}(\theta_{xyz})] \rho_{xyz} - \mathcal{F}(\theta_{xyz}) \alpha_{xyz}, \quad (4.5)$$

with $\mathcal{F}(\theta_{xyz})$ the elasticity of the job finding probability function with respect to market tightness, defined by:

$$\begin{aligned}\mathcal{F}(\theta_{xyz}) &\equiv \theta_{xyz} \frac{p'(\theta_{xyz})}{p(\theta_{xyz})} \\ &= \frac{\theta_{xyh}^\lambda}{1 + \theta_{xyh}^\lambda}.\end{aligned}\tag{4.6}$$

The equilibrium wage (4.5) increases in productivity ρ_{xyz} due to competition between firms, and decrease in the value of amenities α_{xyz} due to competition between workers. But the extent to which these forces matter is governed by tightness in the labor market, through the matching function elasticity $\mathcal{F}(\theta_{xyz}) \in (0, 1)$. The elasticity is larger when there is slack in the market, as firms can exploit their superior bargaining positions to capture a larger part of the surplus. This means that $\mathcal{F}(\theta_{xyz})$ interprets as a measure of firms' bargaining power. Since these different components are functions of hours z , the equilibrium long-hour wage premiums will reflect differences in technology, preferences, and bargaining power.

To better understand the role of market tightness, note that in the slack limit (where $\theta \rightarrow 0$ and $\mathcal{F}(\theta_{xyz}) \rightarrow 1$) firms post wages that only just reward workers for their disutility of employment, at exactly $-\alpha_{xyz}$. When markets become tighter, $\mathcal{F}(\theta_{xyz})$ decreases, as many firms compete for a limited number of workers. This pushes up wages, and in the limit (where $\theta \rightarrow \infty$ and $\mathcal{F}(\theta_{xyz}) \rightarrow 0$) allows workers to capture the entire value of production ρ_{xyz} .⁹ The elasticity of substitution λ governs how responsive bargaining power $\mathcal{F}(\theta_{xyz})$ is to changes in market tightness. In the Leontief limit ($\lambda \rightarrow -\infty$) we have $\mathcal{F}(\theta_{xyz}) = \{0, 1\}$, and wages are set such that the short side of the market captures the entire match surplus. In the Cobb-Douglas limit ($\lambda \rightarrow 0$) we have $\mathcal{F}(\theta_{xyz}) = 0.5$, and wages always split the surplus evenly regardless of the market conditions. At interior values, bargaining power is a smooth function of market tightness.

⁹This assumes throughout that $\rho_{xyz} > -\alpha_{xyz}$ such that there are systematic gains from matching and $\lambda < 0$.

Matching Patterns. We can substitute the equilibrium wage function into the primitives to obtain the following indirect utility and profit functions:

$$\bar{U}_{xyz} = p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz} + \xi_{xyz}, \quad (4.7)$$

$$\bar{\Pi}_{xyz} = q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz} + \eta_{xyz}, \quad (4.8)$$

where $\mathcal{E}_{xyz} = 1 - \mathcal{F}_{xyz}$. Utility and profits thus increase in the fraction of the joint surplus that is captured by workers ($\mathcal{E}_{xyz}\Phi_{xyz}$) and firms ($\mathcal{F}_{xyz}\Phi_{xyz}$) respectively.

The equations (4.7) and (4.8) define a simultaneous two sided discrete choice model, similar to that of [Choo and Siow \(2006\)](#). I follow them in assuming that shocks on both sides are iid Type One Extreme Value distributed.¹⁰ Utility and profit maximization with respect to partner types and hours then implies closed form labor supply ($\mathcal{S}_{yz|x}$) and demand ($\mathcal{D}_{xz|y}$) functions:

$$\mathcal{S}_{yz|x} := \exp(p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz}) a_x, \quad (4.9)$$

$$\text{where: } a_x = \left(\sum_{y,z} \exp(p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz}) \right)^{-1},$$

$$\text{and } \mathcal{D}_{xz|y} := \exp(q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz}) b_y, \quad (4.10)$$

$$\text{where: } b_y = \left(\sum_{x,z} \exp(q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz}) \right)^{-1}.$$

These functions allow us to recover the matching patterns predicted by the model. Let M_{xyz} denote the predicted number of (x, y, z) matches, M_{x0} the number of type x unemployed workers, and M_{0y} the number of type y vacant positions. The equilibrium matching patterns in the model

¹⁰Recent work by [Galichon and Salanié \(2022\)](#) show that this assumption can be relaxed. But since the model is just-identified, we need additional information. The information content of the within-type distribution of transfers is discussed in [Salanié \(2015\)](#), whereas [Fox et al. \(2018\)](#) shows how variation across independent markets can be used to identify the distribution of unobserved heterogeneity. See also related work by [Gualdani and Sinha \(2023\)](#).

are defined by:

$$M_{xyz} = p(\theta_{xyz})\mathcal{S}_{xyz} = q(\theta_{xyz})\mathcal{D}_{xyz}, \quad (4.11)$$

$$M_{x0} = \sum_y (1 - p(\theta_{xyz})) \mathcal{S}_{xyz}, \quad (4.12)$$

$$M_{0y} = \sum_x (1 - q(\theta_{xyz})) \mathcal{D}_{xyz}. \quad (4.13)$$

Characterizing Equilibrium. The following adding-up restrictions hold in equilibrium:

$$\begin{aligned} \sum_{y,z} \mathcal{S}_{xyz} &= n_x \quad \text{for all } x \in X \\ \sum_{x,z} \mathcal{D}_{xyz} &= f_y \quad \text{for all } y \in Y, \end{aligned} \quad (4.14)$$

where n_x denotes the total number of type x workers and f_y the number of type y firms. Together with an equation for market tightness derived from the supply and demand functions, the system obtained by substituting the supply and demand functions into this set of adding-up constraints exhausts the equilibrium conditions of the model:

$$\sum_{y,z} \exp(p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz}) a_x = 1 \quad \text{for all } x \in X, \quad (4.15)$$

$$\sum_{x,z} \exp(q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz}) b_y = 1 \quad \text{for all } y \in Y, \quad (4.16)$$

$$\text{where: } \theta_{xyz} = \exp([q(\theta_{xyz})\mathcal{F}_{xyz} - p(\theta_{xyz})\mathcal{E}_{xyz}]\Phi_{xyz}) \frac{f_y}{n_x} \frac{b_y}{a_x}. \quad (4.17)$$

Theorem 1. Consider the system of equations defined by (4.15)–(4.16) with market tightness implicitly defined by (4.17). Let the job finding probabilities $p(\theta_{xyz})$ follow a Constant Elasticity of Substitution (CES) form as in (4.2). Assume that the joint match surplus Φ_{xyz} and the marginal distributions of worker- and firm types (n_x, f_y) are positive. Suppose that the aggregate substitution effects of the matching prospects (a_x, b_y) of other-type agents are smaller than the effects of the matching prospects of own-type agents as in [Chen et al. \(2021\)](#). Then there exists a unique equilibrium solution (a_x, b_y) to the system.

The proof of Theorem 1 proceeds in two steps. The first step is to show that there exists a

unique solution to the implicit function that defines market tightness (4.17). Given that a unique solution exists, we can substitute it into the mapping defined by (4.15) and (4.16). The second step consists of showing that this mapping is invertible using the results from Chen et al. (2021). To compute the equilibrium solution, we start from an arbitrary guess on market tightness. We can then solve the system defined by (4.15) and (4.16) in (a_x, b_y) . These values allow us to update the guess for market tightness through (4.17). Given the updated guess, we can again solve the system of equations for the equilibrium solution. These steps are iterated until convergence. More details can be found in Appendix A.2.

5 Identification and Estimation

Identification. The main parameters in the model are workers' amenity valuations α_{xyz} , firms' productivity values ρ_{xyz} , and the elasticity of substitution λ . Conditional on the elasticity, identification is similar to the frictionless matching model of Dupuy and Galichon (2022). Labor supply and demand both increase in the joint surplus Φ_{xyz} , so equilibrium matching patterns inform the model about the surplus. The additional observations of equilibrium transfers (monthly wages) allows me to separate out the amenity from the productivity values within the surplus. To see this, note that wages (4.5) increase in productivity ρ_{xyz} but decrease in amenities α_{xyz} .

With observations on wages and matching patterns and wages in a single market, the productivity and amenity parameters exhaust all data moments. Similar to recent work by Chiappori et al. (2017) and Fox et al. (2018), I leverage variation across independent markets $s \in S$ to obtain additional degrees of freedom.¹¹ Assuming that all markets share the same preference structure, I exploit variation in the marginal distributions of worker- and firm types across markets. These compositional differences lead to distinct equilibrium matching patterns and wages. Even with identical preferences, exogenous variation in the marginal distributions of workers and firms shifts tightness, which generates observable differences in outcomes through the nonlinear matching function.¹² Since there is only a single parameter in the matching function, the

¹¹See also Appendix C of Galichon et al. (2019) on identifying the Pareto frontier curvature in matching problems with imperfectly transferable utility. In this model, the elasticity of substitution shapes the curvature.

¹²Several approaches have been proposed to identify additional parameters in similar models. Graham et al. (2014) introduce information about the joint surplus. Dupuy et al. (2023) use measures of firm-side productivity. Salanié (2015) shows how to leverage within-type variation in wages. Ciscato (2024) discusses identification from panel data.

model is identified with just two markets. A proof of identification can be found in Appendix A.3.

Separability. The transparent identifying equations follow from the assumptions about the nature of unobserved heterogeneity. As in Choo and Siow (2006), we assume that worker-side shocks ξ_{xyz} are independent of which particular firm of a given type the worker matches with, and symmetrically that firm-side shocks η_{xyz} are independent of the particular worker of a given type. In other words, we rule out that interactions between unobservables on both sides of the market create additional value. This is a strong assumption, which is mainly addressed by including rich enough types to capture systematic heterogeneity in the joint surplus.

Parameterization. While the model is identified non-parametrically, I introduce a flexible parametric specification for the amenity and productivity functions. Particularly, the value of amenities is modeled as:

$$\alpha_{xyz} = \sum_{x,y} \mathbb{1}_{\alpha_{x \times y}} + \sum_{x,z} \mathbb{1}_{\alpha_{x \times z}}. \quad (5.1)$$

Workers are thus allowed to value amenities differently across occupations, through the fixed effects $\mathbb{1}_{\alpha_{x \times y}}$. This could be related to differences in unobserved amenities or tasks over which preferences diverge. I also allow for workers to differ in their valuation of working hours, through fixed effects $\mathbb{1}_{\alpha_{x \times z}}$.

The occupational production functions are parameterized as:

$$\rho_{xyz} = \sum_{\tilde{x},y} \mathbb{1}_{\rho_{\tilde{x} \times y}} + \sum_{y,z} \mathbb{1}_{\rho_{y \times z}}. \quad (5.2)$$

I allow workers of different ages and education levels ($\tilde{x}$) to differ in their productive value across occupations, as captured by the fixed effects $\mathbb{1}_{\rho_{\tilde{x} \times y}}$. The productive value of different hours options is also allowed to be occupation-specific, as captured by $\mathbb{1}_{\rho_{y \times z}}$.

Data. I use individual level cross-sectional data from several independent and perfectly segregated labor markets. Each market s has information on matching patterns by type, denoted $\hat{M}_{xyz}^s$ for observed (x, y, z) matches, $\hat{M}_{x0}^s$ for unemployed workers of type x , and $\hat{M}_{0y}^s$ for vacancies of type y . I also observe a noisy measure of wages $\hat{W}_{ijk}^s$, assumed to relate to the equilibrium wage

by:

$$\hat{W}_{ijk}^s = W_{xyh}^s + \delta_{ijk}^s \quad \text{with} \quad \delta_{ijk}^s \sim \mathcal{N}(0, \sigma) \quad \text{iid}, \quad (5.3)$$

where δ_{ijk}^s is a centered Gaussian measurement error with standard deviation σ .

Estimation. We estimate the parameters of the model, denoted Ω below, using maximum likelihood. The likelihood function consist of three components. First, the contributions of employed individuals equate predicted matching patterns and wages to their sample counterparts. Second, the contributions of unemployed individuals equate predicted and observed unemployment patterns. Finally, the contributions of vacant positions equate predicted and observed vacancies. Note that we need to integrate out over the unobserved search decisions of unmatched workers and firms. The log likelihood of an individual observation can thus be expressed as:

$$\log \ell_{ijk}^s(\Omega) = \underbrace{\mathbb{1}_{xyz}^s \left[\log M_{xyz|\Omega}^s - \frac{(\hat{W}_{ijk}^s - W_{xyz|\Omega}^s)^2}{2\sigma^2} - \log \sigma \right]}_{\text{Matched: Wage \& Matching}} + \underbrace{\mathbb{1}_{x0}^s \log M_{x0|\Omega}^s}_{\text{Unemployed}} + \underbrace{\mathbb{1}_{0y}^s \log M_{0y|\Omega}^s}_{\text{Vacant}}, \quad (5.4)$$

where $\mathbb{1}_{xyz}^s$ is an indicator that equals one if observation (i, j, k) is a match of type (x, y, z) in market s , $\mathbb{1}_{x0}^s$ is an indicator for observation i being an unemployed person of type x in market s , and $\mathbb{1}_{0y}^s$ is an indicator for observation j being a vacant position of type y in market s . The estimated parameter vector $\hat{\Omega}$ solves:

$$\hat{\Omega} = \arg \max_{\Omega} \left\{ \sum_{s,i,j,k} \log \ell_{ijk}^s(\Omega) \right\}. \quad (5.5)$$

Scaling. The estimation relies on multiple cross-sections of matching pattern, unemployment, and vacancy stocks. But these differ in magnitude from the typical implied steady state hiring and separation flows. To reconcile the large number of matches with low unemployment and vacancy rates, the model would drive up the matching function elasticity λ . As noted earlier, this leads to unstable predictions, where small differences in tightness have large effects on the surplus division. To avoid this, the unemployment stocks are re-scaled, reducing the implied

job-finding rate p^* to 45%.¹³ The stock of vacancies is scaled to match the aggregate job openings to unemployment ratio reported in [Mongey and Horwich \(2023\)](#) based on the [Job Openings and Labor Turnover Survey](#) (JOLTS). This does not alter the likelihood function but scales the inputs. Further details are provided in Appendix [A.4](#).

6 Results

This section first reports the main parameter estimates, emphasizing how different types of workers and firms value hours. I then introduce a decomposition that disentangles how these forces interact with bargaining power to shape the equilibrium long-hour premiums across occupations. Next, I assess the importance of sorting and the wage structure in explaining the gender gap in earnings. Finally, I analyze two counterfactuals aimed at reducing the equilibrium long-hour premium. The first alters firms' long-hour productivity premiums, while the second changes workers' amenity valuations.

6.1 The Value of Hours

The Amenity Value of Hours. Figure [4](#) reports the valuations of hours across worker and occupation types. Panel (a) highlights heterogeneity in how workers value working regular hours, as captured by the $\alpha_{xz'}$ parameters from Equation (5.1).¹⁴ I find that working fewer hours is particularly important for women, who on average value working regular hours at about 60% of their monthly earnings. Younger women value regular hours higher (50 to 75%) than older women (50 to 60% of monthly earnings), while the highest educated group values regular hours lower. Men value regular working hours comparatively less than men, at about 40% of monthly earnings. The main exception are younger men with only a BA degree, whose valuation exceeds 60%.

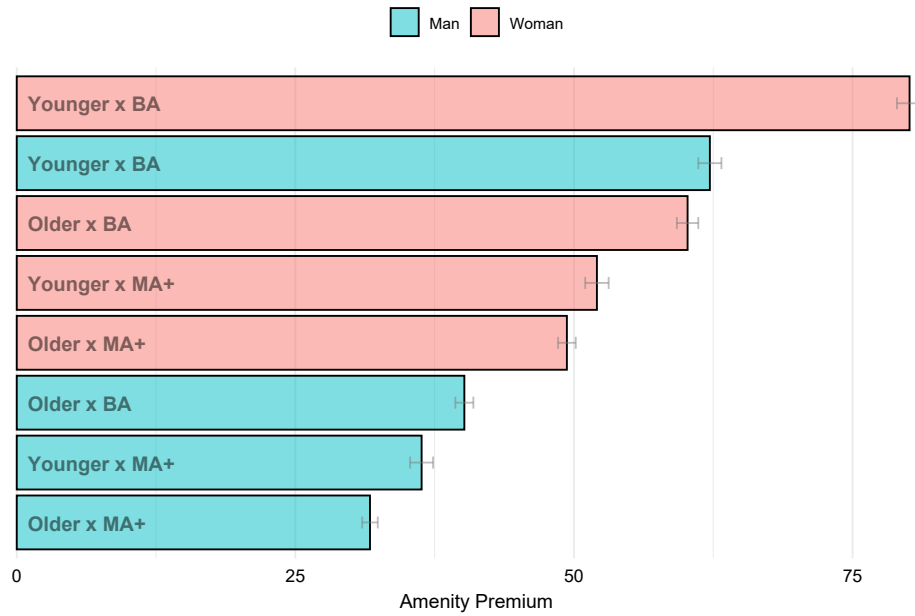
These results are broadly consistent with prior work showing that women value working less hours higher than men, such as [Flabbi and Moro \(2012\)](#), [Wiswall and Zafar \(2018\)](#), and

¹³Job finding rate estimates typically vary between 25 and 45% ([Shimer, 2012](#); [Kudlyak and Lange, 2018](#); [Eeckhout and Sepahsalari, 2024](#)). Because I consider a highly productive and educated sub-sample with a lower than average unemployment rate, who operate in a tight labor market, I choose the upper-range estimate of 45%. The exact choice is relatively inconsequential, and different rates yield comparable estimates.

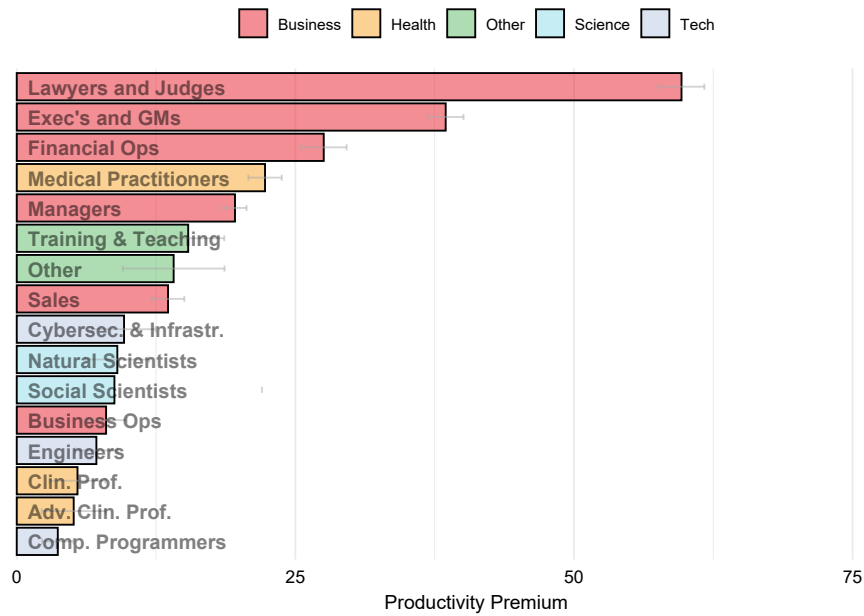
¹⁴It is important to note that we do not know to what extent these valuations reflect preferences, constraints imposed by household responsibilities, or broader societal norms. An overview of different perspectives is provided in [Olivetti et al. \(2024\)](#).

Figure 4: The Value of Hours

(a) Amenity Premiums



(b) Productivity Premiums



Notes. Panel (a) shows the estimated amenity premium for working regular hours across worker types. These values reflect the percentage increase in amenity value from working regular hours, expressed as a percentage of the long hours baseline. Panel (b) shows the estimated productivity premiums for working long hours across occupations. These values reflect the percentage increase in productive value generated by working long hours, expressed as a percentage of the regular hours baseline. Parameters are shown with 95% confidence bounds.

De Schouwer and Kesternich (2024). Because most of the literature focuses on part-time work, direct comparisons are difficult. A notable exception is Maestas et al. (2023), who use a discrete-choice experiment to elicit workers' preferences for long weekly hours. They find that, when hours increase from 40 to 55, which roughly represent regular and long hours in my sample, workers demand earnings premiums just below 40%. While this is somewhat lower than my estimates, this difference may be partly explained by the sample composition. The income effects of a wage change are typically assumed to dominate substitute effects, and I focus on the upper segment of the labor market (Boppart and Krusell, 2020; Mantovani, 2023).

The Productive Value of Hours. Panel (b) reports the firm-side productivity premium from long weekly hours, as captured by the ρ_{yz} parameters in Equation (5.2). The estimated productivity differences generally align with the earnings premiums presented in Figure 2. In Business occupations, such as Lawyers, Judges, Executives, General Managers, and Financial Operations, the large earnings premiums reflect sizable differences in productivity. The additional value created by working long instead of regular hours in these occupations varies between 25 and 60%. In Health occupations, both productive and earnings premiums are generally small, with Medical Practitioners as the main exception.

However, several exceptions caution against equating earnings and productivity premiums. A first example are Tech occupations, where working long hours generates little additional value, but where earnings premiums are nonetheless sizable. This is particularly the case for Computer Scientists, who are estimated to generate less than 5% of additional value when working longer hours, but whose monthly earnings premium is almost 20%. Similarly, in Sales and Business Operations, productive differences are relatively small, yet monthly earnings premiums are substantial. This highlights an important deviation from the narrative of Goldin (2014), and we need to look beyond productivity differences to understand why earnings premiums arise in these occupations.

6.2 Decomposing the Long Hour Premium.

To better understand how equilibrium long-hour premiums are shaped, I introduce a decomposition based on the equilibrium wage function defined in Equation (4.5). Let z denote a long

hour arrangement and z' a regular hour alternative. The long-hour premium across occupations is defined by the expected wage difference between these options $\Delta W_y \equiv \mathbb{E}_x [W_{xyz} - W_{xyz'}]$, which can be decomposed into:

$$\Delta W_y = \mathbb{E}_x \left[\underbrace{[1 - \mathcal{F}(\theta_{yz})](\rho_{yz} - \rho_{yz'})}_{\text{Scaled } \Delta \text{ Productivity}} + \underbrace{\mathcal{F}(\theta_{yz})(\alpha_{yz'} - \alpha_{yz})}_{\text{Scaled } \Delta \text{ Amenity}} + \underbrace{\Phi_{yz'} [\mathcal{F}(\theta_{yz'}) - \mathcal{F}(\theta_{yz})]}_{\Delta \text{ Bargained Surplus}} \right]. \quad (6.1)$$

The equilibrium long-hour premium ΔW_y aggregates three components. The first term captures differences in firms' production technologies, highlighted in Figure 4b. As stressed in Goldin (2014), long-hour premiums increase the more productive working long hours is compared to a regular-hour alternative ($\rho_{yz} > \rho_{yz'}$). Note that, as firms' bargaining power $\mathcal{F}(\theta_{yz})$ increases, they share less of the additional production surplus. The second component captures workers' amenity valuations, highlighted in Figure 4a. Long hour premiums are larger when workers value working regular hours more strongly ($\alpha_{yz'} > \alpha_{yz}$). As firms' bargaining power increases, long-hour premiums are reduced to just rewarding workers for their dislike of long weekly hours. The final term captures a direct bargaining power effect. At a given match surplus ($\Phi_{yz'}$), the earnings premium increases when long-hour markets are tighter. In other words, when the ratio of firms posting long hour vacancies over workers to supply long hours ($\theta_{xyz'}$) is larger than the corresponding ratio for regular hours (θ_{xyz}), the long-hour premium increases.

Results. The decomposition results are presented in Figure 5. The first column highlights the contributions of the scaled difference in workers' valuations of long weekly hours (Scaled Δ Amenity). In some occupations, and particularly those for whom the estimated productive differences are small, long-hour premiums simply reflect a compensation for workers' dislike of working additional hours. Some examples are Advanced Clinical Professionals, Scientists, and Teaching & Training occupations, where the amenity difference explains the entire earnings premium. In fact, in these occupations the earnings premium is smaller than the scaled amenity difference, because of the relative oversupply of long-hour workers. Recent work by Gicheva (2020) offers similar observations, showing that workers are more willing to work long hours at lower wage

premiums in occupations with higher social value.¹⁵

Figure 5: Decomposition of Premium

	Amenity	Productive	Power	
Lawyers and Judges	61.85	37.43	0.71	Business
Exec's and GMs	67.99	34.12	-2.1	
Financial Ops	78.39	23.19	-1.58	
Managers	74.42	22.47	3.11	
Sales	75.88	17.48	6.64	
Business Ops	91.39	8	0.61	
Medical Practitioners	77.05	27.13	-4.19	Health
Clin. Prof.	94.08	4.73	1.19	
Adv. Clin. Prof.	114.4	3.76	-18.16	
Cybersec. & Infrastr.	92.48	9.87	-2.34	Tech
Engineers	87.8	8.93	3.27	
Comp. Programmers	85.42	6.48	8.1	
Natural Scientists	97.97	8.42	-6.39	Science
Social Scientists	107.97	6.99	-14.96	
Other	105.08	9.63	-14.71	Other
Training & Teaching	101.47	9.05	-10.52	

Notes. This figure highlights the decomposition of the long-hour premium across occupations shown in Equation (6.1), expressed as a percentage of the total earnings difference. The first column shows the Scaled Δ Amenity term, the second column shows the Scaled Δ Productivity term, and the final column shows the Δ Bargained Surplus term.

The second column shows the scaled difference in productive values (Scaled Δ Productivity). In line with the earlier results, productive differences are particularly important in Business occupations, where they explain up to 40% of the earnings premium, but less so in others. To better understand how long-hour premiums are determined, the third column shows the contribution of bargaining power (Δ Bargained Surplus). As noted before, this is important in reducing the long-hour premium of Advanced Clinical Professionals, Scientists, and in Teaching & Training occupations, which decrease by 10 to 20%. In these occupations, they account for a substantially larger fraction of the long-hour premium compared to productive differences. In other occupations, there is a relative under-supply of long-hour workers, so the bargaining term pushes up

¹⁵This could be tested through the model by further relaxing the parametric structure of the amenity component in Equation (5.1), but at the cost of an additional $Y \times Z$ parameters.

the earnings premiums. The main examples are Computer Programmers and Sales occupations, where excess demand in long-hour positions increases the premiums by 6 to 8%.

6.3 The Gender Gap in Earnings

The next step in the analysis is to study the implications for gender differences in earnings. To do so, I compute the earnings gap in the model, defined by:

$$G \equiv \sum_{y,h} \mathcal{M}_{xyz} W_{xyz} - \sum_{y,h} \mathcal{M}_{x'yz} W_{x'yz}, \quad (6.2)$$

where $\mathcal{M}_{xyz}$ denotes the normalized mass of men x (women x') in occupation y and hours option z within the model. I derive a decomposition similar to that of [Oaxaca \(1973\)](#) and [Blinder \(1973\)](#), separating the effects of men and women sorting differently across hours and occupations from the wage structure component. Substituting the equilibrium wage function into the expression, I provide further insights into what shapes the wage rates, similar to [Morchio and Moser \(2024\)](#):

$$G = \sum_{y,z} \left(\underbrace{\Delta \mathcal{M}_y \frac{\mathcal{M}_{x'yz}}{\mathcal{M}_{x'y}}}_{\Delta \text{ Sorting in } y} + \underbrace{\mathcal{M}_{xy} \Delta \mathcal{M}_{z|y}}_{\Delta \text{ Sorting in } z} \right) W_{x'yz} + \sum_{y,z} \mathcal{M}_{xyz} \left(\underbrace{[1 - \mathcal{F}(\theta_{xyz})] \Delta \rho_{yz}}_{\Delta \text{ Productivity}} + \underbrace{\mathcal{F}(\theta_{xyz}) \Delta \alpha_{yz}}_{\Delta \text{ Preferences}} + \underbrace{\Phi_{x'yz} \Delta \mathcal{F}_{yz}}_{\Delta \text{ Bargaining}} \right). \quad (6.3)$$

See Appendix [A.5](#) for the derivation.

Results. The results from this decomposition are presented in Table [4](#). The first panel highlights that the gender earnings gap in the data is 26.2%, while the model predicts a gap of 17.4%. This means that about a third of the earnings difference is not explained by the model. An important reason for this discrepancy is that gender does not directly enter the production function, which rules out firms' ability to directly discriminate in the wages they offer men and women of the same type. While estimates on discrimination and productive differences vary significantly, 36% falls well within the range of values commonly ascribed to discrimination or (perceived) productivity differences ([Flabbi, 2010](#); [Blau and Kahn, 2017](#); [Olivetti et al., 2024](#))

Table 4: Wage Gap Decomposition

	Percent
<i>Model vs. Data</i>	
Data Gap	26.2%
Model Gap	17.4%
Unexplained	36.8%
<i>Decomposition</i>	
Sorting: Occupations	34.8%
Sorting: Hours	11.4%
Earnings: Amenity	36.6%
Earnings: Bargaining	-19.8%

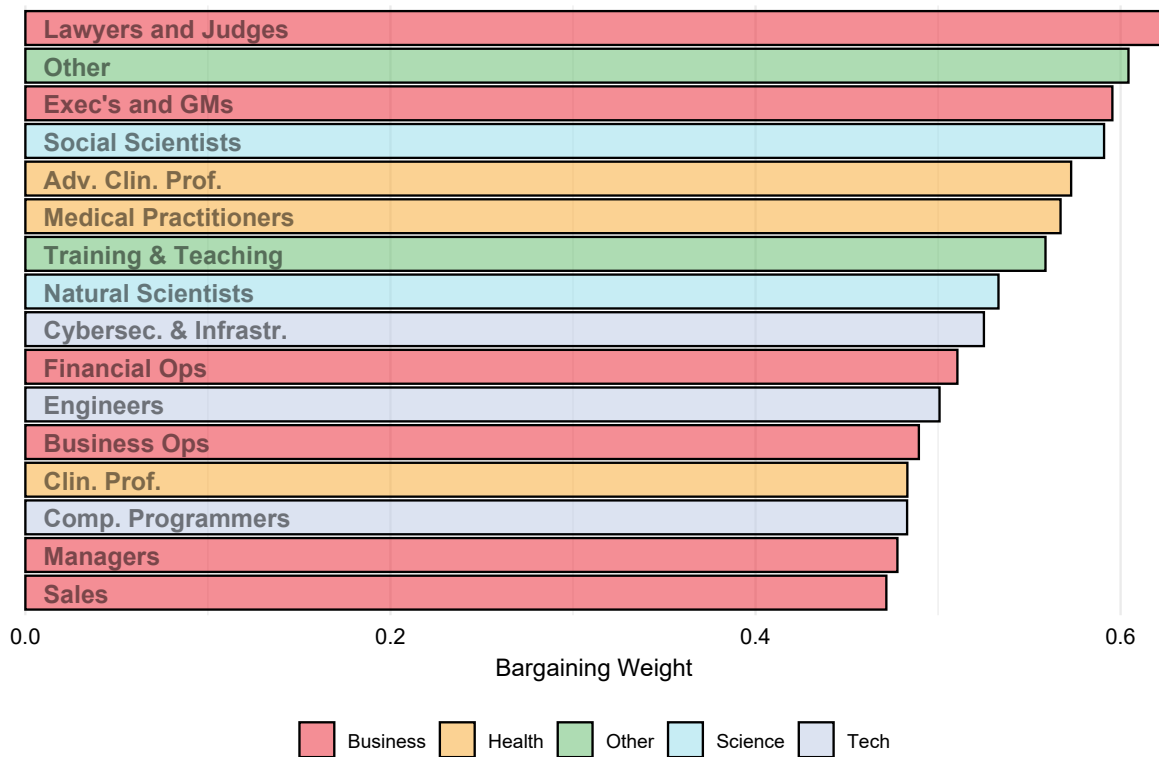
Notes. This table reports the gender wage gap in the data and model. The model's gap is decomposed into contributions reflecting differential sorting and differences in the wage structure, as defined in equation (6.3). The contributions of these different components are expressed as percentages of the total gap.

The second panel highlights how the different components shape the earnings gap. Sorting into different occupations and hours is an important determinant of the gender difference in earnings. If women worked in the same occupations as men, the earnings gap would decrease by about 35%. This is very close to [Blau and Kahn \(2017\)](#), who find that 33% of the hourly wage gap stems from occupational sorting, with similar estimates being reported in [Olivetti et al. \(2024\)](#). Differential sorting into regular hours explains a smaller, but still sizable 11% of the earnings gap. This is again very close to the numbers reported in [Olivetti et al. \(2024\)](#), who find that about 10% of the earnings gap stems from differences in hours worked.

Differences in how compensation is structured for men and women also explain an important part of the gender gap in earnings. There is a substantial gender gap in the value of amenities, and equalizing this difference would decrease the earnings gap by about 37%. This falls in the middle range of recent estimates, with [Maestas et al. \(2023\)](#) finding that amenities explain about a quarter of the compensation gap, while [Morchio and Moser \(2024\)](#) and [Bell \(2024\)](#) find that amenities explain between 50 and 66%. Finally, I also find an important offsetting contribution by bargaining power. This reflects that women sort into tighter markets than men, reducing the gender wage gap by almost 20%.

Bargaining Power. To better understand the gender difference in bargaining power, Figure 6 shows the occupational distribution of firms' bargaining weights. An important reason for the disparity is that women are more likely to work in Clinical Professional and Business Operations occupations, where they make up 83% and 55% of the workforce, respectively. In these occupations, which are characterized by excess demand, firm bargaining weights are below 0.5, increasing workers' wages. Conversely, occupations with the largest relative excesses in supply, as reflected in firm bargaining weights around 0.6, are all predominantly male. The difference is particularly large in Executive, GM, and Social Scientist occupations, where men make up more than 65% of the workforce. These results are consistent with the broader labor-market trend that women are more likely to be employed in service industries (Ngai and Petrongolo, 2017; Albanesi and Şahin, 2018), which are characterized by tighter labor markets (Şahin et al., 2014).

Figure 6: Firms' Bargaining Weights



Notes. This figure highlights the distribution of bargaining power across occupations as captured by the median bargaining weights of firms ($\mathcal{F}_{yz}$).

6.4 Counterfactual Simulations

Reducing the Productivity Premium. I now study the effect of policies aimed at increasing gender equality by reducing the long-hour earnings premium. The first reduces firms' productivity gains from working long weekly hours. As Goldin (2014) notes, the gender gap may be considerably reduced if firms did not have strong incentives to reward long and particular hours of work highly. To simulate this, I reduce the long-hour productivity premiums ρ_{yz} by some rate τ , and compute the counterfactual equilibrium wages and matching patterns for these alternative production structures.

Table 5 shows the results from this analysis, with the rows representing the outcomes at differently scaled long-hour premiums $\tau\rho_{yz}$. This highlights that policies aimed at changing the production structure are relatively ineffective at reducing the gender gap in earnings. A complete removal of the productivity gain ($\tau = 0$) only reduces the gender earnings gap by about 5%. The contributions of the different components highlight that re-sorting is the main reason for this reduction. Differential re-sorting across occupations reduces the gap by about 2.3%, while resorting into regular hour arrangements reduces the gap by about 5%. These changes are relatively small, as not only men, but also women are sorting out of long-hour arrangements.

Table 5: Counterfactuals: Production

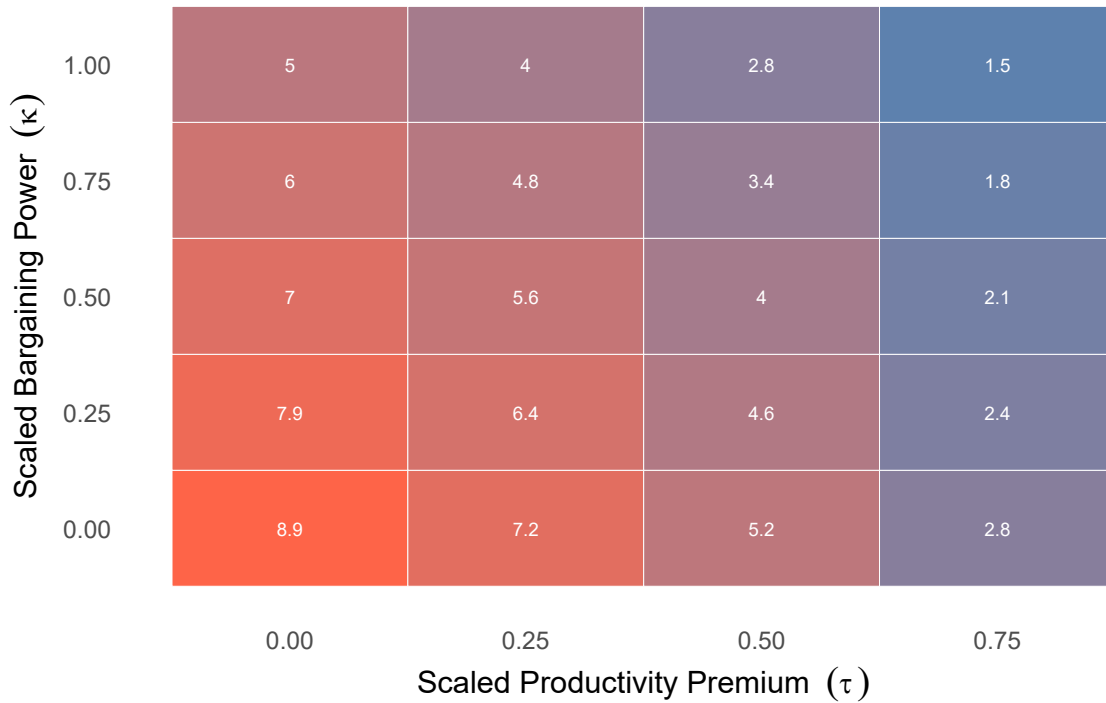
τ	Model Gap	$\Delta G/G_0$	S_y/G_0	S_z/G_0	$\Delta W_{\mathcal{F}}/G_0$	$\Delta W_{\alpha}/G_0$
1.00	17.39%	0.00%	–	–	–	–
0.75	17.13%	-1.47%	-0.81%	-1.27%	-0.20%	0.73%
0.50	16.90%	-2.80%	-1.44%	-2.55%	-0.38%	1.43%
0.25	16.69%	-3.99%	-1.94%	-3.80%	-0.54%	2.08%
0.00	16.51%	-5.04%	-2.34%	-4.99%	-0.66%	2.69%

Notes. This table reports the gender gap in the model at the counterfactual gaps at different reductions τ of the long-hour productivity premium. The changes are further decomposed into sorting and wage structure changes, as defined as in Equation (6.3). The different components are expressed as a percentage change from the baseline earnings gap G_0 .

The effect of the policy is reduced by women re-sorting into better amenity jobs, which offsets about 2.7% of the earnings gap decrease. As highlighted in 4a, women value regular hour jobs with better amenities substantially higher than men. When women leave long-hour positions, they move into higher amenity regular-hour jobs, but they pay a substantial compensating

differential to do so. When workers have stronger bargaining positions, the amenity valuations are less reflected in equilibrium wages, which enhances the effectiveness of these policies. Figure 7 illustrates this point. For each τ , I shift bargaining power exogenously by scaling down firm weights $\mathcal{F}_{xyz}$ by factor κ . The first row ($\kappa = 1$) shows the baseline effect, where fully reducing the productivity premium leads to a 5% reduction of the gender wage gap. In the final row ($\kappa = 0$), wages only reflect firm-side productive differences. In this case, the effectiveness of a policy that removes the productivity premium almost doubles, reducing the gender gap by almost 9%.

Figure 7: Counterfactuals: Production and Power



Notes. This figure highlights the distribution of bargaining power across occupations as captured by the median bargaining weights of firms ($\mathcal{F}_{yz}$).

Equalizing Amenity Valuations. The next counterfactual scenario equalizes men and women's amenity valuations of hours. This could reflect changes induced by a policy that expands child-care or parental leave, alleviating household burdens and reducing women's effective costs of working long hours. I construct counterfactual scenario's that blend together men and women's

preferences over hours at a rate $\tau \in [0, 1]$:

$$\alpha_{xz}^\tau = (1 - \tau)\alpha_{xz} + \tau\alpha_{x'z}. \quad (6.4)$$

At each τ I again re-solve the model to obtain matching patterns and wages, holding all other primitives fixed. We then compute the earnings gap in the counterfactual equilibrium (G_τ) and study how it differs from the baseline gap. Changes in the earnings gap (ΔG) are again decomposed into contributions that stem from re-sorting and changes in the equilibrium wage structure.

Table 6: Counterfactual: Preferences

τ	Model Gap	$\Delta G/G_0$	S_y/G_0	S_z/G_0	$\Delta W_{\mathcal{F}}/G_0$	$\Delta W_\alpha/G_0$
0.00	17.39%	0.00%	–	–	–	–
0.25	15.37%	-11.62%	-0.47%	-4.99%	3.53%	-9.73%
0.50	13.36%	-23.13%	-0.96%	-9.75%	6.91%	-19.40%
0.75	11.39%	-34.50%	-1.48%	-14.24%	10.13%	-29.02%
1.00	9.45%	-45.67%	-2.03%	-18.41%	13.21%	-38.58%

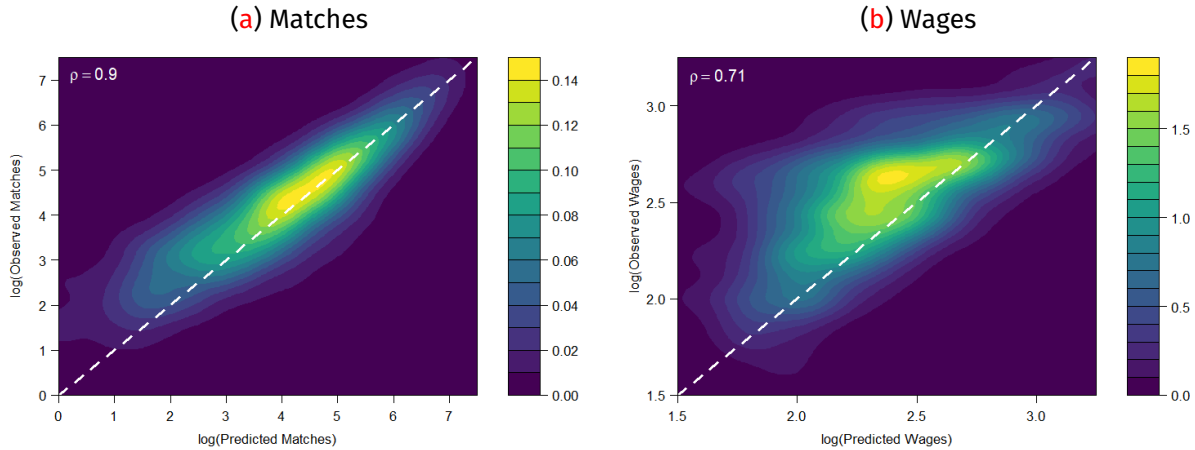
Notes. This table reports the gender gap in the model at the counterfactual gaps at different convergence rates τ of the amenity valuations of men and women. The model wage gap is decomposed into sorting and wage change components, as defined as in equation (6.3).

Table 6 reports the counterfactual gaps and decomposition results. As the convergence rate τ increases, the gender gap falls from about 19.3% to below 9.5% when amenity valuations are equalized, a reduction of more than 45%. The first main contribution comes from men and women re-sorting into jobs with more similar hours, reducing the earnings gap by roughly 20%, while occupational re-sorting contributes just 2%. The bulk of the earnings gap reduction stems from differences in the wage structure. Because women's valuations of regular hours fall while men's rise, the compensating differentials they are willing to pay converge. At equal valuations, the earnings gap is reduced by almost 40%. Importantly, because they are sorting into more similar hours options and occupations, women lose their bargaining power advantage, raising the earnings gap by over 10%.

6.5 Model Fit and Further Estimates

Figure 8 highlights the model fit by plotting the sample matching patterns and wages in each sub-market ($X \times Y \times Z \times S$) against their predicted model counterparts. The diagonal line indicates that the model perfectly fits the data. Panel (a) shows that the model replicates matching patterns very well, with a correlation between observed and predicted values of 0.90. Note that this means that hours in the model align closely with those in the data. The main markets where the model does not fit well are those with few observations, for which the model under-predicts the number of matches.

Figure 8: Model Fit



Notes. This figure plots the model predictions against the data moments. Panel (a) highlights the matching pattern fit, and Panel (b) the fit of wages. The 45° line indicates that the model exactly replicates the data. Both matching patterns and wages are expressed in logs. The color scale highlights the density of observations obtained from a kernel estimator.

Panel (b) shows that the model also does a good job replicating wages, with a correlation of 0.71. In general, the model under-predicts wages slightly, and struggles mainly with replicating upper-intermediate level wages. Appendix C.1 focuses in particular on how the model fits occupational long-hour premiums, which are generally well-replicated. Because the model generates less dispersion, it over-estimates the long-hour premium in some occupations, notably those where wages are set less competitively, being Health and Science. In terms of explaining individual wages, the model has an $R^2 = 0.181$. This is comparable to what a wage regressions with only worker- and occupation type fixed effects explains in our sample, which has $R^2 = 0.189$.

Further Estimates. The remaining parameter can be found in Appendix C. Note that the elasticity of substitution λ estimate is -0.69 (s.e. 0.004). This is a higher than the value in, for example, [Eeckhout and Sepahsalari \(2024\)](#) who set it at 0.41. But this difference mainly stems from the fact that we study a sample for which we assumed a higher job finding rate. The Appendix also presents workers' preferences over occupation types, and productive heterogeneity across occupations for different types of workers

7 Conclusion

This paper studies how hours and wages are determined in the labor market, and why the earnings premium for long weekly hours varies across occupations. I first empirically document that hours, wages, and equilibrium long-hour premiums all respond to changes in labor market tightness. To better understand this relation, I introduce a tractable equilibrium search and matching model, which features directed search as in [Moen \(1997\)](#), but also allows for two sided multidimensional heterogeneity as in [Choo and Siow \(2006\)](#). The model highlights that occupational long-hour wage premiums are not only shaped by production technologies, but also by workers' amenity valuations and tightness in the labor market.

While previous work has emphasized the importance of productive heterogeneity in shaping long-hour premiums, I highlight the importance of amenities and market tightness. In line with the narrative of [Goldin \(2014\)](#), productive differences are indeed important in shaping the large long-hour premiums in highly paid Business occupations. Similarly, in other occupations, notably in Health, both productive and earnings premiums are small. But there are several exceptions, where productive and earnings differences are misaligned. Some examples are Computer Programmers or Sales occupations, in which we observe relatively large earnings premiums, which are partly inflated by tightness in the long-hour market. In other occupations, notably those with high social value, such as Health and Science, there is an excess supply of long-hour workers, reducing the earnings premiums by 10 to 20%.

The next step considers the implications for the labor market. In line with recent estimates, I find that men and women sorting into jobs in different occupations at different hours explains about half of the earnings difference. I also provide further insights into the contributions of

the wage structure. Particularly, another 33% of the earnings gap stems from men and women's amenity values. I also document an interesting structural advantage for women, who sort into tighter markets where they have more bargaining power. Absent this difference the gender gap would increase by about 20%. I then study how changing the worker- and firm side valuations of hours affects the gender wage gap. In a counterfactual simulation, I show that removing the productivity premium to long weekly hours does little to change the gender gap, but becomes more effective in tight markets where women pay lower compensating differentials for re-sorting into regular hour jobs. An alternative simulation that aims to converge men and women's preferences over hours reduces the gap by almost 50%.

To conclude, I point out several directions for future research. A first extension is to make the model dynamic, which would particularly yield further insights into supply-side motivations. While previous work by [Jang and Yum \(2022\)](#) has highlighted the importance of human capital accumulation and dynamic returns to long hours, these dimensions are not captured in the model. Another extension of the model would be to introduce households, as labor supply choices are clearly not made in isolation. An interesting feature of such a model is that it would allow us to study how the intra-household bargaining process is shaped by the structure of the labor market.

References

- Aaronson, Daniel and Eric French (2004) “The effect of part-time work on wages: Evidence from the social security rules,” *Journal of Labor Economics*, 22 (2), 329–352.
- Acemoglu, Daron, David Autor, Jonathon Hazell, and Pascual Restrepo (2022) “Artificial intelligence and jobs: Evidence from online vacancies,” *Journal of Labor Economics*, 40 (S1), S293–S340.
- Albanesi, Stefania and Ayşegül Şahin (2018) “The gender unemployment gap,” *Review of Economic Dynamics*, 30, 47–67.
- Andresen, Martin Eckhoff and Emily Nix (2025) “You Can’t Force Me Into Caregiving: Paternity Leave and the Child Penalty,” *The Economic Journal*, ueaf057.
- Arcidiacono, Peter, Andrew Beauchamp, and Marjorie McElroy (2016) “Terms of endearment: An equilibrium model of sex and matching,” *Quantitative Economics*, 7 (1), 117–156.
- Autor, David, Arindrajit Dube, and Annie McGrew (2023) “The unexpected compression: Competition at work in the low wage labor market,” Technical report, National Bureau of Economic Research.
- Azar, José, Ioana Marinescu, and Marshall Steinbaum (2022) “Labor market concentration,” *Journal of Human Resources*, 57 (S), S167–S199.
- Beauchamp, Andrew, Rossella Calvi, and Scott Fulford (2024) “Terms of Engagement: Migration, Dowry, and Love in Indian Marriages.”
- Bell, Alex (2024) “Job Amenities & Earnings Inequality.”
- Bertrand, Marianne, Claudia Goldin, and Lawrence F Katz (2010) “Dynamics of the gender gap for young professionals in the financial and corporate sectors,” *American economic journal: applied economics*, 2 (3), 228–255.
- Bick, Alexander, Adam Blandin, and Richard Rogerson (2022) “Hours and wages,” *The Quarterly Journal of Economics*, 137 (3), 1901–1962.

- Blau, Francine D and Lawrence M Kahn (2017) “The gender wage gap: Extent, trends, and explanations,” *Journal of economic literature*, 55 (3), 789–865.
- Blinder, Alan S (1973) “Wage discrimination: reduced form and structural estimates,” *Journal of Human resources*, 436–455.
- Bonhomme, Stéphane and Grégory Jolivet (2009) “The pervasive absence of compensating differentials,” *Journal of Applied Econometrics*, 24 (5), 763–795.
- Boppart, Timo and Per Krusell (2020) “Labor supply in the past, present, and future: a balanced-growth perspective,” *Journal of Political Economy*, 128 (1), 118–157.
- Börschlein, Erik-Benjamin, Mario Bossler, and Martin Popp (2024) “Scarce Workers, High Wages?” *arXiv preprint arXiv:2408.04508*.
- Bureau of Labor Statistics (2024) “Fastest Growing Occupations,” <https://www.bls.gov/ooh/fastest-growing.htm>, Accessed: 2025-04-09.
- Chade, Hector, Jan Eeckhout, and Lones Smith (2017) “Sorting through search and matching models in economics,” *Journal of Economic Literature*, 55 (2), 493–544.
- Chen, Liang, Eugene Choo, Alfred Galichon, and Simon Weber (2021) “Matching function equilibria with partial assignment: Existence, uniqueness and estimation,” *arXiv preprint arXiv:2102.02071*.
- Chiappori, Pierre-André, Monica Costa Dias, and Costas Meghir (2018) “The marriage market, labor supply, and education choice,” *Journal of Political Economy*, 126 (S1), S26–S72.
- (2020) “Changes in assortative matching: Theory and evidence for the US,” Technical report, National Bureau of Economic Research.
- Chiappori, Pierre-André, Carlo V Fiorio, Alfred Galichon, and Stefano Verzillo (2025) “Assortative matching on income,” Technical report, JRC Working Papers in Economics and Finance.
- Chiappori, Pierre-André, Bernard Salanié, and Yoram Weiss (2017) “Partner choice, investment in children, and the marital college premium,” *American Economic Review*, 107 (8), 2109–2167.

- Choo, Eugene (2015) "Dynamic marriage matching: An empirical framework," *Econometrica*, 83 (4), 1373–1423.
- Choo, Eugene and Aloysius Siow (2006) "Who marries whom and why," *Journal of political Economy*, 114 (1), 175–201.
- Ciscato, Edoardo (2024) "Assessing racial and educational segmentation in large marriage markets."
- Ciscato, Edoardo and Simon Weber (2020) "The role of evolving marital preferences in growing income inequality," *Journal of Population Economics*, 33 (1), 307–347.
- Corblet, Pauline (2023) "Education Expansion, Sorting, and the Decreasing Wage Premium."
- Cortés, Patricia and Jessica Pan (2019) "When time binds: Substitutes for household production, returns to working long hours, and the skilled gender wage gap," *Journal of Labor Economics*, 37 (2), 351–398.
- De Schouwer, Thimo and Iris Kesternich (2024) "Work meaning and the flexibility puzzle," *Journal of Labor Economics*, forthcoming.
- Deming, David J and Kadeem Noray (2020) "Earnings dynamics, changing job skills, and STEM careers," *The Quarterly Journal of Economics*, 135 (4), 1965–2005.
- Diamond, Peter A (1982) "Wage determination and efficiency in search equilibrium," *The Review of Economic Studies*, 49 (2), 217–227.
- Dupuy, Arnaud and Alfred Galichon (2014) "Personality traits and the marriage market," *Journal of Political Economy*, 122 (6), 1271–1319.
- (2022) "A note on the estimation of job amenities and labor productivity," *Quantitative Economics*, 13 (1), 153–177.
- Dupuy, Arnaud, Alfred Galichon, Sonia Jaffe, and Scott Duke Kominers (2020) "Taxation in matching markets," *International Economic Review*, 61 (4), 1591–1634.
- Dupuy, Arnaud, John Kennes, and Ran Sun Lyng (2023) "Job Amenities in the Market for CEOs."

- Eeckhout, Jan and Philipp Kircher (2010) "Sorting and decentralized price competition," *Econometrica*, 78 (2), 539–574.
- Eeckhout, Jan and Alireza Sepahsalari (2024) "The effect of wealth on worker productivity," *Review of Economic Studies*, 91 (3), 1584–1633.
- Erosa, Andrés, Luisa Fuster, Gueorgui Kambourov, and Richard Rogerson (2022) "Hours, occupations, and gender differences in labor market outcomes," *American Economic Journal: Macroeconomics*, 14 (3), 543–590.
- (2024) "Wage and earnings inequality between and within occupations: The role of labor supply," *Journal of Monetary Economics*, 103553.
- Flabbi, Luca (2010) "Gender discrimination estimation in a search model with matching and bargaining," *International Economic Review*, 51 (3), 745–783.
- Flabbi, Luca and Andrea Moro (2012) "The effect of job flexibility on female labor market outcomes: Estimates from a search and bargaining model," *Journal of Econometrics*, 168 (1), 81–95.
- Fox, Jeremy T, Chenyu Yang, and David H Hsu (2018) "Unobserved heterogeneity in matching games," *Journal of Political Economy*, 126 (4), 1339–1373.
- Galichon, Alfred, Scott Duke Kominers, and Simon Weber (2019) "Costly concessions: An empirical framework for matching with imperfectly transferable utility," *Journal of Political Economy*, 127 (6), 2875–2925.
- Galichon, Alfred and Bernard Salanié (2022) "Cupid's invisible hand: Social surplus and identification in matching models," *The Review of Economic Studies*, 89 (5), 2600–2629.
- Gicheva, Dora (2020) "Occupational social value and returns to long hours," *Economica*, 87 (347), 682–712.
- Goldin, Claudia (2014) "A grand gender convergence: Its last chapter," *American economic review*, 104 (4), 1091–1119.
- Goldin, Claudia and Lawrence F Katz (2016) "A most egalitarian profession: pharmacy and the evolution of a family-friendly occupation," *Journal of Labor Economics*, 34 (3), 705–746.

- Graham, Bryan S, Guido W Imbens, and Geert Ridder (2014) "Complementarity and aggregate implications of assortative matching: A nonparametric analysis," *Quantitative Economics*, 5 (1), 29–66.
- Greenwood, Jeremy, Nezih Guner, Georgi Kocharkov, and Cezar Santos (2016) "Technology and the changing family: A unified model of marriage, divorce, educational attainment, and married female labor-force participation," *American Economic Journal: Macroeconomics*, 8 (1), 1–41.
- Gualdani, Cristina and Shruti Sinha (2023) "Partial identification in matching models for the marriage market," *Journal of Political Economy*, 131 (5), 1109–1171.
- Hagedorn, Marcus and Iourii Manovskii (2008) "The cyclical behavior of equilibrium unemployment and vacancies revisited," *American Economic Review*, 98 (4), 1692–1706.
- Hershbein, Brad and Lisa B Kahn (2018) "Do recessions accelerate routine-biased technological change? Evidence from vacancy postings," *American Economic Review*, 108 (7), 1737–1772.
- Hoeck, Christian Philip (2022) "Wage effects of labor market tightness," Technical report, Danmarks Nationalbank Working Papers.
- Hwang, Hae-shin, Dale T Mortensen, and W Robert Reed (1998) "Hedonic wages and labor market search," *Journal of Labor Economics*, 16 (4), 815–847.
- Jaffe, Sonia and Simon Weber (2019) "The effect of meeting rates on matching outcomes," *Economic Theory*, 67 (2), 363–378.
- Jang, Youngsoo and Minchul Yum (2022) "Nonlinear occupations and female labor supply over time," *Review of Economic Dynamics*, 46, 51–73.
- Kudlyak, Marianna and Fabian Lange (2018) "Measuring Heterogeneity in Job Finding Rates among the Non-Employed Using Labor Force Status Histories."
- Lamadon, Thibaut, Jeremy Lise, Costas Meghir, and Jean-Marc Robin (2024) "Labor market matching, wages, and amenities," Technical report, National Bureau of Economic Research.
- Lang, Kevin and Sumon Majumdar (2004) "The pricing of job characteristics when markets do not clear: theory and policy implications," *International Economic Review*, 45 (4), 1111–1128.

- Lavetti, Kurt and Ian M Schmutte (2025) “Estimating Compensating Wage Differentials with Endogenous Job Mobility.”
- Legros, Patrick and Andrew F Newman (2007) “Beauty is a beast, frog is a prince: Assortative matching with nontransferabilities,” *Econometrica*, 75 (4), 1073–1102.
- Liu, Kai (2016) “Explaining the gender wage gap: Estimates from a dynamic model of job changes and hours changes,” *Quantitative Economics*, 7 (2), 411–447.
- Macpherson, David A and Barry T Hirsch (1995) “Wages and gender composition: why do women’s jobs pay less?” *Journal of labor Economics*, 13 (3), 426–471.
- Maestas, Nicole, Kathleen J Mullen, David Powell, Till Von Wachter, and Jeffrey B Wenger (2023) “The value of working conditions in the United States and implications for the structure of wages,” *American Economic Review*, 113 (7), 2007–2047.
- Manning, Alan and Barbara Petrongolo (2008) “The part-time pay penalty for women in Britain,” *The economic journal*, 118 (526), F28–F51.
- Mantovani, Cristiano (2023) “Hours-Biased Technological Change,” Technical report, Working Paper.
- McFadden, Daniel (1978) “Modelling the choice of residential location,” *Transportation Research Record*, 672, 72–77.
- Menzio, Guido and Shouyong Shi (2010) “Block recursive equilibria for stochastic models of search on the job,” *Journal of Economic Theory*, 145 (4), 1453–1494.
- Moen, Espen R (1997) “Competitive search equilibrium,” *Journal of political Economy*, 105 (2), 385–411.
- Mongey, Simon and Jeff Horwich (2023) “Are Job Vacancies Still as Plentiful as They Appear? Implications for the ‘Soft Landing’,” *Blog, Federal Reserve Bank of Minneapolis*.
- Morchio, Iacopo and Christian Moser (2024) “The gender pay gap: Micro sources and macro consequences,” Technical report, National Bureau of Economic Research.

- Mortensen, Dale T (1982) "The matching process as a noncooperative bargaining game," in *The economics of information and uncertainty*, 233–258: University of Chicago Press.
- Mourifié, Ismael (2019) "A marriage matching function with flexible spillover and substitution patterns," *Economic Theory*, 67 (2), 421–461.
- Mourifié, Ismael and Aloysius Siow (2021) "The Cobb-Douglas marriage matching function: Marriage matching with peer and scale effects," *Journal of Labor Economics*, 39 (S1), S239–S274.
- Ngai, L Rachel and Barbara Petrongolo (2017) "Gender gaps and the rise of the service economy," *American Economic Journal: Macroeconomics*, 9 (4), 1–44.
- Oaxaca, Ronald (1973) "Male-female wage differentials in urban labor markets," *International economic review*, 693–709.
- Olivetti, Claudia, Jessica Pan, and Barbara Petrongolo (2024) "The evolution of gender in the labor market," in *Handbook of labor economics*, 5, 619–677: Elsevier.
- Peters, Michael (1984) "Bertrand equilibrium with capacity constraints and restricted mobility," *Econometrica: Journal of the Econometric Society*, 1117–1127.
- Pissarides, Christopher A (1985) "Short-run equilibrium dynamics of unemployment, vacancies, and real wages," *The American Economic Review*, 75 (4), 676–690.
- Reynoso, Ana (2024) "The impact of divorce laws on the equilibrium in the marriage market," *Journal of Political Economy*, 132 (12), 4155–4204.
- Rosen, Sherwin (1986) "The theory of equalizing differences," *Handbook of labor economics*, 1, 641–692.
- Şahin, Ayşegül, Joseph Song, Giorgio Topa, and Giovanni L Violante (2014) "Mismatch unemployment," *American Economic Review*, 104 (11), 3529–3564.
- Salanié, Bernard (2015) "Identification in separable matching with observed transfers," *Report*. [154, 161] *Sattinger, M.(1979), "Differential rents and the distribution of earnings." Oxford Economic Papers*, 31 (1), 60–71.

- San, Shmuel (2024) "Who Works Where and Why: The Role of Social Connections in the Labor Market."
- Shimer, Robert (2012) "Reassessing the ins and outs of unemployment," *Review of Economic Dynamics*, 15 (2), 127–148.
- Siow, Aloysius (2015) "Testing Becker's theory of positive assortative matching," *Journal of Labor Economics*, 33 (2), 409–441.
- Sorkin, Isaac (2018) "Ranking firms using revealed preference," *The quarterly journal of economics*, 133 (3), 1331–1393.
- Sullivan, Paul and Ted To (2014) "Search and nonwage job characteristics," *Journal of Human Resources*, 49 (2), 472–507.
- Taber, Christopher and Rune Vejlin (2020) "Estimation of a roy/search/compensating differential model of the labor market," *Econometrica*, 88 (3), 1031–1069.
- Wiswall, Matthew and Basit Zafar (2018) "Preference for the workplace, investment in human capital, and gender," *The Quarterly Journal of Economics*, 133 (1), 457–507.
- Wright, Randall, Philipp Kircher, Benoît Julien, and Veronica Guerrieri (2021) "Directed search and competitive search equilibrium: A guided tour," *Journal of Economic Literature*, 59 (1), 90–148.
- Yurdagul, Emircan (2017) "Production complementarities and flexibility in a model of entrepreneurship," *Journal of Monetary Economics*, 86, 36–51.

Appendix

A Model

A.1 Equilibrium Tightness and Wages

I start by writing the equilibrium problem similar to [Wright et al. \(2021\)](#), as firms maximizing profits subject to offering workers their expected equilibrium utilities:

$$\begin{aligned} \max_{W, \theta, x, z} \quad & q(\theta_{xyz})(\rho_{xyz} - W_{xyz}) + \eta_{xyz} \\ \text{s.t.} \quad & u_{xyz} = p(\theta_{xyz})(\alpha_{xyz} + W_{xyz}). \end{aligned} \tag{A.1}$$

Rewriting the constraint in terms of wages and substituting it out of the objective function yields:

$$\max_{W, \theta} \left\{ \frac{p(\theta_{xyz})(\rho_{xyz} + \alpha_{xyz}) - u_{xyz} + \eta_{xyz}\theta_{xyz}}{\theta_{xyz}} \right\}. \tag{A.2}$$

Taking the first order condition of (A.2) with respect to market tightness (θ_{xyz}) we get:

$$u_{xyz} = (\rho_{xyz} + \alpha_{xyz}) (p(\theta_{xyz}^*) - \theta_{xyz}^* p'(\theta_{xyz}^*)). \tag{A.3}$$

Substituting this back into the constraint yields a characterization of equilibrium wages:

$$W_{xyz}^* = (1 - \mathcal{F}(\theta_{xyz}^*)) \rho_{xyz} - \mathcal{F}(\theta_{xyz}^*) \alpha_{xyz}, \tag{A.4}$$

with $\mathcal{F}(\theta_{xyz}^*)$ the elasticity of the job finding probability function with respect to market tightness:

$$\mathcal{F}(\theta_{xyz}^*) := \theta_{xyz}^* \frac{p'(\theta_{xyz}^*)}{p(\theta_{xyz}^*)}. \tag{A.5}$$

I use expression (A.4) to substitute out wages and obtain the indirect utility (4.7) and profit (4.8) functions shown in section 4.2.

A.2 Existence and Uniqueness of Equilibrium

I now establish the existence of a unique equilibrium in the model. Starting from equations (4.15)–(4.17), I substitute out $p(\theta_{xyz})$ and $q(\theta_{xyz})$ using the matching function defined in equation (4.2), as well as $\mathcal{E}_{xyz}$ and $\mathcal{F}_{xyz}$ using the definition from equation (4.6). This yields the following system of $X + Y$ equations that characterizes the equilibrium:

$$\sum_{y, z} \exp \left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) a_x = 1 \quad \text{for all } x, \quad (\text{A.6})$$

$$\sum_{x, z} \exp \left(\theta_{xyz}^{\lambda-1} (1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) b_y = 1 \quad \text{for all } y, \quad (\text{A.7})$$

$$\text{where: } \theta_{xyz} = \exp \left(\frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz} \right) \frac{a_x f_y}{b_y n_x}. \quad (\text{A.8})$$

I want to show that there exists a unique equilibrium solution (a_x, b_y) to this system. Suppose that $\lambda < 0$, $\Phi_{xyz} > 0$, $n_x > 0$, and $f_y > 0$. Then both a_x and b_y take on values in $(0, 1)$. I proceed by first showing that the implicit function (A.8) that defines market tightness has a unique solution for any given (a_x, b_y) to ensure that the system is well-defined. I then establish uniqueness of the equilibrium itself, as defined by the system of equations (A.6) and (A.7), using an argument similar to [Chen et al. \(2021\)](#).

Existence of a Unique Root. Market tightness is strictly positive, because it measures the number of searching firms over searching workers, so I look for a root on $(0, +\infty)$ of the function:

$$G(\theta_{xyz}) = \theta_{xyz} - \exp \left(\frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz} \right) \frac{a_x f_y}{b_y n_x}. \quad (\text{A.9})$$

At the limits, this expression yields:

$$\lim_{\theta_{xyz} \rightarrow 0} G(\theta_{xyz}) = - \exp(\Phi_{xyz}) \frac{a_x f_y}{b_y n_x},$$

$$< 0,$$

$$\lim_{\theta_{xyz} \rightarrow \infty} G(\theta_{xyz}) = \theta_{xyz} - \exp(-\Phi_{xyz}) \frac{a_x f_y}{b_y n_x},$$

$$= \infty.$$

Since $G(\cdot)$ is continuous, the intermediate value theorem guarantees that a root exists. To show uniqueness, I examine the derivative of $G(\cdot)$. Rewriting the inner exponent term as:

$$\psi(\theta_{xyz}) := \frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz}, \quad (\text{A.10})$$

the derivative writes:

$$G'(\theta_{xyz}) = 1 - \exp(\psi(\theta_{xyz})) \psi'(\theta_{xyz}) \frac{a_x}{b_y} \frac{f_y}{n_x}. \quad (\text{A.11})$$

Since all multiplicative terms are strictly positive, the sign of $G'(\theta_{xyz})$ depends on $\psi'(\theta_{xyz})$:

$$\psi'(\theta_{xyz}) = \Phi_{xyz} \frac{(\lambda - 1)(\theta_{xyz} + 1)\theta_{xyz}^{\lambda-2}}{(1 + \theta_{xyz}^\lambda)^{(2\lambda-1)/\lambda}} < 0. \quad (\text{A.12})$$

This term is strictly negative because $\Phi_{xyz} > 0$, $\lambda < 0$, and $\theta_{xyz} > 0$. It follows that $G'(\theta_{xyz}) > 1$, and thus that $G(\theta_{xyz})$ is strictly increasing and has a unique root. In turn, this implies that there exists a unique solution θ_{xyz}^* that solves (A.8) for every (x, y, z) .

Existence of a Unique Equilibrium. Consider now the system defined by (A.6) and (A.7), which is a mapping $T : (a_x, b_y) \mapsto (a'_x, b'_y)$, with $\theta_{xyz}(a_x, b_y)$ defined implicitly by (A.15):

$$a'_x = \left(\sum_{y,z} \exp \left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) \right)^{-1} \quad \text{for all } x, \quad (\text{A.13})$$

$$b'_y = \left(\sum_{x,z} \exp \left(\theta_{xyz}^{\lambda-1} (1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) \right)^{-1} \quad \text{for all } y, \quad (\text{A.14})$$

$$\text{with: } \theta_{xyz}(a_x, b_y) = \exp \left(\frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz} \right) \frac{a_x}{b_y} \frac{f_y}{n_x}. \quad (\text{A.15})$$

Given the previous assumptions, the surplus within the exponential is multiplied by a fraction in $(0, 1)$, so we can consider the compact domain $\mathcal{K} \subset (0, 1)^{X+Y}$ given by:

$$\underline{a}_x = \frac{1}{\sum_{y,z} \exp(\Phi_{xyz})} \leq a_x \leq \frac{1}{Y \times Z} = \bar{a}_x, \quad (\text{A.16})$$

$$\underline{b}_y = \frac{1}{\sum_{x,z} \exp(\Phi_{xyz})} \leq b_y \leq \frac{1}{X \times Z} = \bar{b}_y. \quad (\text{A.17})$$

By construction, the functions (A.13)–(A.14) map $\mathcal{K}$ into itself, so $T(\mathcal{K}) \subseteq \mathcal{K}$. An equilibrium exists by Brouwer's fixed point theorem.

To prove uniqueness, I follow the approach of [Chen et al. \(2021\)](#). Taking logs of the system:

$$\begin{aligned}\sigma_x(a_x, b_y) &:= \log a_x + \log \sum_{y,z} \exp(A(\theta_{xyz}(a_x, b_y))), \\ \sigma_y(a_x, b_y) &:= \log b_y + \log \sum_{x,z} \exp(B(\theta_{xyz}(a_x, b_y))),\end{aligned}$$

where the functions $A(\theta_{xyz})$ and $B(\theta_{xyz})$ are the inner exponent terms in the a'_x and b'_y equations (A.13)–(A.14) respectively. The equilibrium conditions are equivalent to $\sigma(a, b) = 0$.

The Jacobian has the following structure:

$$\begin{aligned}\frac{\partial \sigma_x}{\partial a_x} &= \frac{1}{a_x} + \frac{\sum_{y,z} \exp(A(\theta_{xyz})) A'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial a_x}}{\sum_{y,z} \exp(A(\theta_{xyz}))}, \\ \frac{\partial \sigma_x}{\partial a_{x'}} &= 0 \quad \text{for } x' \neq x, \\ \frac{\partial \sigma_x}{\partial b_y} &= \frac{\sum_z \exp(A(\theta_{xyz})) A'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial b_y}}{\sum_{y,z} \exp(A(\theta_{xyz}))}, \\ \frac{\partial \sigma_y}{\partial b_y} &= \frac{1}{b_y} + \frac{\sum_{x,z} \exp(B(\theta_{xyz})) B'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial b_y}}{\sum_{x,z} \exp(B(\theta_{xyz}))}, \\ \frac{\partial \sigma_y}{\partial b_{y'}} &= 0 \quad \text{for } y' \neq y, \\ \frac{\partial \sigma_y}{\partial a_x} &= \frac{\sum_z \exp(B(\theta_{xyz})) B'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial a_x}}{\sum_{x,z} \exp(B(\theta_{xyz}))},\end{aligned}$$

where we can show that $A'(\theta) > 0$, $B'(\theta) < 0$, and by the implicit function theorem $\frac{\partial \theta_{xyz}}{\partial a_x} > 0$ and $\frac{\partial \theta_{xyz}}{\partial b_y} < 0$. So we have that

$$\frac{\partial \sigma_x}{\partial a_x} > 0, \quad \frac{\partial \sigma_x}{\partial b_y} \leq 0, \quad \frac{\partial \sigma_x}{\partial a_{x'}} = 0, \tag{A.18}$$

$$\frac{\partial \sigma_y}{\partial b_y} > 0, \quad \frac{\partial \sigma_y}{\partial a_x} \leq 0, \quad \frac{\partial \sigma_y}{\partial b_{y'}} = 0, \tag{A.19}$$

These (in)equalities imply the weak substitutes condition of [Chen et al. \(2021\)](#). We then follow

them in assuming that the following *diagonal dominance* property holds:

$$\left| \frac{\partial \sigma_x}{\partial a_x} \right| > \sum_y \left| \frac{\partial \sigma_x}{\partial b_y} \right| \quad \text{for all } x, \quad (\text{A.20})$$

$$\left| \frac{\partial \sigma_y}{\partial b_y} \right| > \sum_x \left| \frac{\partial \sigma_y}{\partial a_x} \right| \quad \text{for all } y, \quad (\text{A.21})$$

where it should be noted that the terms $\frac{\partial \sigma_x}{\partial a_{x'}} = \frac{\partial \sigma_y}{\partial b_{y'}} = 0$. These conditions are sufficient for the mapping to be injective, as argued in [Chen et al. \(2021\)](#). This proves that the equilibrium exists and is unique.

A.3 Identification

I now discuss the identification of the productivity (ρ_{xyz}), amenity (α_{xyz}) and matching function elasticity (λ) parameters. Assume that the sample consists of matching patterns and wages in multiple segmented markets $s \in S$, with the additional superscripts omitted for clarity. I start by expressing the observed matching patterns and wages in terms of their nonlinear model expressions, and then show that these equations are invertible in the structural parameters. This establishes identification, assuming that the data contains a sufficient number of markets.

The model expressions are given by:

$$\begin{aligned}\hat{M}_{xyz}/\hat{n}_x &= \mathcal{S}_{yz|x}p(\theta_{xyz}), \\ &= \exp\left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) a_x(1 + \theta_{xyz}^\lambda)^{1/\lambda},\end{aligned}\tag{A.22}$$

$$\begin{aligned}\hat{M}_{xyz}/\hat{f}_y &= \mathcal{D}_{xz|y}q(\theta_{xyz}), \\ &= \exp\left(\theta_{xyz}^{\lambda-1}(1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) b_y(1 + \theta_{xyz}^\lambda)^{1/\lambda}\theta_{xyz}^{-1},\end{aligned}\tag{A.23}$$

$$\begin{aligned}\hat{M}_{x0}/\hat{n}_x &= \sum_y \mathcal{S}_{yz|x}(1 - p(\theta_{xyz})), \\ &= \sum_y \exp\left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) a_x \left(1 - (1 + \theta_{xyz}^\lambda)^{1/\lambda}\right),\end{aligned}\tag{A.24}$$

$$\begin{aligned}\hat{M}_{0y}/\hat{f}_y &= \sum_x \mathcal{D}_{xz|y}(1 - q(\theta_{xyz})), \\ &= \sum_x \exp\left(\theta_{xyz}^{\lambda-1}(1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) b_y \left(1 - (1 + \theta_{xyz}^\lambda)^{1/\lambda}\theta_{xyz}^{-1}\right),\end{aligned}\tag{A.25}$$

$$\hat{W}_{xyz} = \frac{\rho_{xyz} - \theta_{xyz}^\lambda \alpha_{xyz}}{1 + \theta_{xyz}^\lambda}.\tag{A.26}$$

Suppose that $\lambda < 0$ to ensure that the job- and worker finding probability functions are well-defined, and that $\Phi_{xyz} > 0$ such that there are gains to be made from matching. I further only include types for which there are observations, so both n_x and f_y are strictly positive, as is market tightness θ_{xyz} . Note that we do not observe θ_{xyz} , because the search decisions of unmatched

agents are unknown. But for a given parameter vector, the model yields a unique θ_{xyz} up to type specific scales (a_x, b_y) , which are pinned down from the observation of unmatched agents. So I establish identification of the structural parameters, recognizing that θ_{xyz} is itself an equilibrium object determined by these parameters and the observed marginals.

Amenities and Productivity. The joint surplus $\Phi_{xyz} := \alpha_{xyz} + \rho_{xyz}$ can be identified from the observed number of matches $\hat{M}_{xyz}$. In both (A.22) and (A.23), all terms are strictly positive on the relevant domain, and Φ_{xyz} enters linearly in the exponent. Although Φ_{xyz} also influences market tightness θ_{xyz} , implicitly differentiating the tightness equation verifies that θ_{xyz} is strictly increasing in Φ_{xyz} . Recall that θ_{xyz} is implicitly defined by:

$$\theta_{xyz} = \exp [\psi(\theta_{xyz}) \Phi_{xyz}] C_{xy}, \quad (\text{A.27})$$

where $\psi(\theta_{xyz})$ is a function of θ_{xyz} and C_{xy} a positive constant. Differentiating both sides with respect to Φ_{xyz} yields:

$$\frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} = \frac{\partial \exp [\psi(\theta_{xyz}) \Phi_{xyz}] C_{xy}}{\partial \Phi_{xyz}} \quad (\text{A.28})$$

$$= \exp [\psi(\theta_{xyz}) \Phi_{xyz}] C_{xy} \left[\psi(\theta_{xyz}) + \Phi_{xyz} \psi'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} \right] \quad (\text{A.29})$$

Substituting in $\theta_{xyz} = \exp [\psi(\theta_{xyz}) \Phi_{xyz}] C_{xy}$ and rewriting:

$$\frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} = \theta_{xyz} \left(\psi(\theta_{xyz}) + \Phi_{xyz} \psi'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} \right) \quad (\text{A.30})$$

$$= \frac{\psi(\theta_{xyz}) \theta_{xyz}}{1 - \Phi_{xyz} \psi'(\theta_{xyz}) \theta_{xyz}} > 0. \quad (\text{A.31})$$

This expression is strictly positive since $\psi(\theta_{xyz}) > 0$ and $\psi'(\theta_{xyz}) < 0$ for all $\lambda < 0$ and $\theta_{xyz} > 0$. Therefore, market tightness is strictly increasing in surplus Φ_{xyz} , so the number of matches increases monotonically in the surplus, allowing us to invert these expressions to identify Φ_{xyz} .

Separate identification of the amenity and productivity terms that make up the surplus follows from the observation of wages. As shown in equation (A.26), equilibrium wages increase in productivity (ρ_{xyz}) and decrease in amenities (α_{xyz}) at a given value of tightness. Since θ_{xyz}

is determined by the equilibrium conditions for a given surplus, the wage equation provides a one-to-one mapping from $(\rho_{xyz}, \alpha_{xyz})$ to observed wages.

The Elasticity of Substitution. The parameter λ governs the curvature of the matching function, and is identified from variation across markets. I assume that the surplus terms ρ_{xyz} and α_{xyz} are invariant across markets $s \in S$, but market tightness $\theta_{xyz|s}$ varies due to differences in the marginal worker ($n_{x|s}$) and firm ($f_{y|s}$) distributions. Equation (A.26) shows that equilibrium wages are also a nonlinear function of λ . Differentiating this expression yields:

$$\frac{\partial \hat{W}_{xyz|s}}{\partial \lambda} = -\frac{\theta_{xyz|s}^\lambda \log(\theta_{xyz|s})(\alpha_{xyz} + \rho_{xyz})}{(1 + \theta_{xyz|s}^\lambda)^2}. \quad (\text{A.32})$$

The sign of this derivative will depend on whether $\theta_{xyz|s} > 1$ or $\theta_{xyz|s} < 1$. But for any fixed $\theta_{xyz|s} > 0$, this function is strictly monotonic in λ . Assuming that the data contains multiple markets with information on wages and matching patterns for varying marginals in a sub-market (x, y, z) , this system can be inverted to recover λ . Since the model contains only a single elasticity parameter, observing $S = 2$ distinct markets suffices for identification.

A.4 Steady State Scaling

Consider a textbook discrete time [Diamond \(1982\)](#), [Mortensen \(1982\)](#) & [Pissarides \(1985\)](#) search model. Let u_t and v_t denote the unemployment and vacancy rates at time t , and let market tightness be defines as $\theta_t = \frac{v_t}{u_t}$. Matches form at flow rate:

$$M_t = p(\theta_t) u_t, \quad (\text{A.33})$$

where $p(\theta_t)$ is the job finding rate and $q(\theta_t) = \frac{p(\theta_t)}{\theta_t}$ the vacancy filling rate. Workers separate from employment at a rate δ , so unemployment evolves according to:

$$u_{t+1} = \delta(1 - u_t) + [1 - p(\theta_t)] u_t. \quad (\text{A.34})$$

In steady state $u_{t+1} = u$ and $\theta_t = \theta$, so unemployment and flow matches are given by:

$$u = \frac{\delta}{\delta + p(\theta)} \quad \text{and} \quad M = p(\theta)u. \quad (\text{A.35})$$

This implies that unemployment and vacancies are:

$$u = \frac{M}{p(\theta)} \quad \text{and} \quad v = \frac{M}{q(\theta)}. \quad (\text{A.36})$$

Scaling. In cross-sectional data only stocks of matches ($\hat{M}_{xyz}$), unemployed workers ($\hat{M}_{x0}$), and vacant jobs ($\hat{M}_{y0}$) are observed. I aggregate these into:

$$E = \sum_{x,y,z} \hat{M}_{xyz} \quad \text{and} \quad U = \sum_x \hat{M}_{x0} \quad \text{and} \quad V = \sum_y \hat{M}_{y0}. \quad (\text{A.37})$$

In steady state, the flow of matches equals the flow of separations:

$$M = \delta E. \quad (\text{A.38})$$

Using raw stocks without adjustment would overstate the matching efficiency by roughly $\frac{\delta+p}{p}$. This forces the estimated matching-elasticity λ to highly negative values, which leads to volatile

model predictions. To avoid this, I scale the data to match plausible steady-state job-finding and hiring rates. Empirical estimates of the job finding rate tend to vary between 25 and 45% (Shimer, 2012; Kudlyak and Lange, 2018; Eeckhout and Sepahsalari, 2024). I set the rate at 45%, in the upper range. I do so because I study a highly productive and educated sub-sample, with a lower than average unemployment rate, who operate in a tight labor market. The implied vacancy filling rate $q^* = p^*/\theta$ is set to match the average vacancies per unemployed worker reported in Mongey and Horwich (2023) based on Job Openings and Labor Turnover Survey (JOLTS).

I introduce scaled stocks as:

$$U^* = k_n U \quad \text{and} \quad V^* = k_f V, \quad (\text{A.39})$$

and impose:

$$p^* = M/U^* \quad \text{and} \quad q^* = M/V^*. \quad (\text{A.40})$$

Solving for the scaling factors we get:

$$k_n = \frac{\delta E}{p^* U} \quad \text{and} \quad k_f = \frac{\delta E}{q^* V}. \quad (\text{A.41})$$

The k_n and k_f scaled inputs are used instead of the raw stocks when solving the model and likelihood. To calculate standard errors, I compute the Hessian on the raw data at the optimum $\hat{\Omega}$ estimated from the scaled data.

A.5 Decomposition

Let $\mathcal{M}_{xyz}$ be the normalized mass matches of individuals of type x in occupation types y and hours arrangements z , and W_{xyz} their wages. We can write the gender wage gap G as:

$$G \equiv \sum_{y,z} \mathcal{M}_{xyz} W_{xyz} - \sum_{y,z} \mathcal{M}_{x'yz} W_{x'yz}. \quad (\text{A.42})$$

We can decompose the wage gap into two components, a sorting component and a wage structure component. We can rewrite the gender gap by adding and subtracting $\sum_{y,z} \mathcal{M}_{xyz} W_{x'yz}$ to obtain:

$$G = \underbrace{\sum_{y,z} \Delta \mathcal{M}_{yz} W_{x'yz}}_{\Delta \text{ Sorting}} + \underbrace{\sum_{y,z} \Delta W_{yz} \mathcal{M}_{xyz}}_{\Delta \text{ Wages}},$$

where $\Delta \mathcal{M}_{yz} = \mathcal{M}_{xyz} - \mathcal{M}_{x'yz}$ and $\Delta W_{yz} = W_{xyz} - W_{x'yz}$.

Δ Sorting. Some further algebra allows us to decompose the first term into an occupational sorting component and an hours components within occupations. Let $\mathcal{M}_{xy} = \sum_z \mathcal{M}_{xyz}$ and $\mathcal{M}_{x'y} = \sum_z \mathcal{M}_{x'yz}$ be the occupational margins. We can rewrite $\Delta \mathcal{M}_{yz}$ as:

$$\Delta \mathcal{M}_{yz} = \Delta \mathcal{M}_y \frac{\mathcal{M}_{x'yz}}{\mathcal{M}_{x'y}} + \mathcal{M}_{xy} \Delta \mathcal{M}_{z|y}, \quad (\text{A.43})$$

where $\mathcal{M}_y = \mathcal{M}_{xy} - \mathcal{M}_{x'y}$ and $\Delta \mathcal{M}_{z|y} = \left(\frac{\mathcal{M}_{xyz}}{\mathcal{M}_{xy}} - \frac{\mathcal{M}_{x'yz}}{\mathcal{M}_{x'y}} \right)$. The sorting term writes:

$$\Delta \text{ Sorting} = \sum_{y,z} \left(\underbrace{\Delta \mathcal{M}_y \frac{\mathcal{M}_{x'yz}}{\mathcal{M}_{x'y}}}_{\Delta \text{ Sorting across } y} + \underbrace{\mathcal{M}_{xy} \Delta \mathcal{M}_{z|y}}_{\Delta \text{ Sorting across } z} \right) W_{x'yz}. \quad (\text{A.44})$$

Δ Wages. We can substitute the equilibrium wage function (4.5) into the wage structure component and rewrite the wage difference ΔW_{yz} as:

$$\Delta W_{yz} = [1 - \mathcal{F}(\theta_{xyz})] \Delta \rho_{yz} + \mathcal{F}(\theta_{xyz}) \Delta \alpha_{yz} + \Phi_{x'yz} \Delta \mathcal{F}(\theta_{yz'}). \quad (\text{A.45})$$

Decomposition. Substituting the expressions for sorting and wage differences into (A.42), the gender wage gap can be written as the sum of five terms, as in Equation (6.3) in Section 6.3:

$$\begin{aligned}
 G = & \sum_{y,z} \left(\underbrace{\Delta \mathcal{M}_y \frac{\mathcal{M}_{x'yz}}{\mathcal{M}_{x'y}}}_{\Delta \text{ Sorting in } y} + \underbrace{\mathcal{M}_{xy} \Delta \mathcal{M}_{z|y}}_{\Delta \text{ Sorting in } z} \right) W_{x'yz} + \\
 & \sum_{y,z} \mathcal{M}_{xyz} \left(\underbrace{[1 - \mathcal{F}(\theta_{xyz})] \Delta \rho_{yz}}_{\Delta \text{ Productivity}} + \underbrace{\mathcal{F}(\theta_{xyz}) \Delta \alpha_{yz}}_{\Delta \text{ Preferences}} + \underbrace{\Phi_{x'yz} \Delta \mathcal{F}_{yz}}_{\Delta \text{ Bargaining}} \right),
 \end{aligned} \tag{A.46}$$

where $\Delta \rho_{yz} = \rho_{xyz} - \rho_{x'yz}$, $\Delta \alpha_{yz} = \alpha_{x'yz} - \alpha_{xyz}$, and $\Delta \mathcal{F}_{yz} = \mathcal{F}(\theta_{x'yz}) - \mathcal{F}(\theta_{xyz})$.

B Data

B.1 Occupation Types

This Appendix provides an overview of the individual occupations, as defined by their three digit SOC codes, that are included within each occupation type. Note that the individual occupations are the same as those selected in [Goldin \(2014\)](#).

Business

Business Ops

Buyers and Purchasing Agents, Claims Adjusters, Appraisers, Examiners, and Investigators, Compensation, Benefits, and Job Analysis Specialists, Compliance Officers, Cost Estimators, Human Resources Workers, Logisticians and Project Management Specialists, Management Analysts, Market Research Analysts and Marketing Specialists, Meeting, Convention, and Event Planners, Miscellaneous Business Operations Specialists, Public Relations Specialists

Exec's and GMs

Chief Executives, General and Operations Managers

Financial Ops

Accountants and Auditors, Actuaries, Budget Analysts, Credit Analysts, Credit Counselors and Loan Officers, Financial Analysts and Advisors, Financial Examiners, Miscellaneous Financial Specialists

Lawyers and Judges

Judges, Magistrates, and Other Judicial Workers, Lawyers and Judicial Law Clerks

Sales

Advertising Sales Agents, First-Line Supervisors Non-Retail Sales, Insurance Sales Agents, Miscellaneous Sales Representatives, Services, Sales Engineers, Sales Representatives, Wholesale and Manufacturing, Sales and Related Workers, All Other, Securities, Commodities, and Financial Services Sales Agents

Managers

Administrative Services and Facilities Managers, Advertising and Promotions Managers, Architectural and Engineering Managers, Compensation and Benefits Managers, Computer and Information Systems Managers, Construction Managers, Economists, Emergency Management Directors, Financial Managers, Human Resources Managers, Industrial Production Managers, Marketing and Sales Managers, Medical and Health Services Managers, Miscellaneous Managers, Public Relations and Fundraising Managers, Purchasing Managers, Training and Development Managers, Transportation, Storage, and Distribution Managers

Health

Adv. Clin. Prof.

Nurse Anesthetists, Nurse Practitioners, Physician Assistants, Veterinarians

Clin. Prof.

Audiologists, Dietitians and Nutritionists, Miscellaneous Healthcare Diagnosing or Treating Practitioners, Psychologists, Registered Nurses, Therapists

Medical Practitioners

Chiropractors, Dentists, Optometrists, Pharmacists, Physicians, Podiatrists, Surgeons

Other

Training & Post-Secondary Teachers

Education and Childcare Administrators, Post-Secondary Teachers, Training and Development Specialists

Other

Actors, Producers, and Directors, Agents and Business Managers of Artists, Performers, and Athletes, Air Traffic Controllers and Airfield Operations Specialists, Aircraft Pilots and Flight Engineers, Detectives and Criminal Investigators, First-Line Supervisors of Firefighting and Prevention Workers, First-Line Supervisors of Law Enforcement Workers, Ship and Boat Captains and Operators, Writers and Editors,

Science

Natural Scientists

Agricultural and Food Scientists, Astronomers and Physicists, Atmospheric and Space Scientists,

Biological Scientists, Chemists and Materials Scientists, Environmental Scientists and Geoscientists, Miscellaneous Life Scientists, Miscellaneous Mathematical Science Occupations, Miscellaneous Physical Scientists, Natural Sciences Managers, Operations Research Analysts

Social Scientists

Architects, Except Naval, Miscellaneous Social Scientists and Related Workers, Surveyors, Cartographers, and Photogrammetrists, Urban and Regional Planners

Tech

Comp. Programmers

Computer and Information Analysts, Software Developers and Computer Programmers

Cybersec. & Infrastr.

Database and Network Administrators and Architects, Information Security Analysts, Miscellaneous Computer Occupations

Engineers

Aerospace Engineers, Bioengineers and Biomedical Engineers, Chemical Engineers, Civil Engineers, Computer Hardware Engineers, Electrical and Electronics Engineers, Environmental Engineers, Industrial Engineers, Including Health and Safety, Marine Engineers and Naval Architects, Materials Engineers, Mechanical Engineers, Mining, Petroleum, and Geological Engineers, Miscellaneous Engineers

B.2 Hours within Types

This Appendix highlights variation in average working hours within two types of hours across occupations. In full-time positions, the average number of hours is roughly 40, which is comparable across occupations. In long-hour positions, average hours are about 55, which is again similar across occupations. The main exception is Medical Practitioners, who work about 60 hours per week in a long hour position.

Table 7: Hours within Types

Occupation Type	Regular Hours	Long Hours
Adv. Clin. Prof.	40.25	55.11
Business Ops	40.60	54.11
Clin. Prof.	39.36	56.25
Comp. Programmers	40.42	54.45
Cybersec. & Infrastr.	40.48	54.56
Engineers	40.81	54.31
Exec's and GMs	41.19	55.26
Financial Ops	40.64	54.61
Lawyers and Judges	40.73	55.58
Managers	40.88	54.24
Medical Practitioners	40.37	60.46
Natural Scientists	40.45	54.78
Other	40.48	56.75
Sales	40.78	54.78
Social Scientists	40.68	54.38
Training & Teaching	40.49	55.27

Notes. Summary statistics from the latest [American Community Survey](#) (ACS) wave 2023. The selected sample consists of higher educated workers in high earning degrees as introduced in [Goldin \(2014\)](#) and discussed in section 3.

B.3 The Vacancy Data

This appendix provides additional details on how to construct and scale the distributions of job vacancies over occupations for each state and year. We rely on two sources of data, which respectively inform us about the total number of vacancies and the occupational distributions.

1. *Vacancies: Aggregate Levels.* The [Job Openings and Labor Turnover Survey](#) (JOLTS) is collected by the Bureau of Labor Statistics (BLS) in the United States. The survey questions more than 20,000 establishments about their job openings and hires each month. Based on these questions, the BLS estimates the total number of job openings in the United States. For a vacancy to be measured in JOLTS, three conditions need to be satisfied on the last business day of the month: (i) a position exists and has work available, (ii) the job can start within thirty days, and (iii) there is active recruiting for workers from outside of the establishment. The BLS also reports the number of job openings *per unemployed worker*, which allows for simple scaling with our sample. [Mongey and Horwich \(2023\)](#) adjust their numbers to better align with other macroeconomic series.
2. *Vacancies: Occupational Distribution.* The [National Labor Exchange](#) (NLx) Research Hub job vacancy database contains information on individual job postings in the United States. These vacancies identify several characteristics of the job. We use information on the timing, region of employment, and occupation to construct the regional distributions of occupational vacancies. We discuss more features of this data in section [B.4](#).

B.4 The National Labor Exchange (NLx) Research Hub Data

The National Labor Exchange (NLx) is a non-profit online labor exchange operated by the [National Association of State Workforce Agencies](#) (NASWA), which represents the workforce agencies of all states in the United States, and by the [DirectEmployers Association](#) (DE). The exchange aggregates job vacancies from various individual sources, such as companies' career websites, states' job vacancy banks, and the federal jobs portal. The vacancies are vetted to omit junk records and remove duplicates. The Research Hub of the NLx provides data – both real-time and historical snapshots – about all the job vacancies available on the exchange.

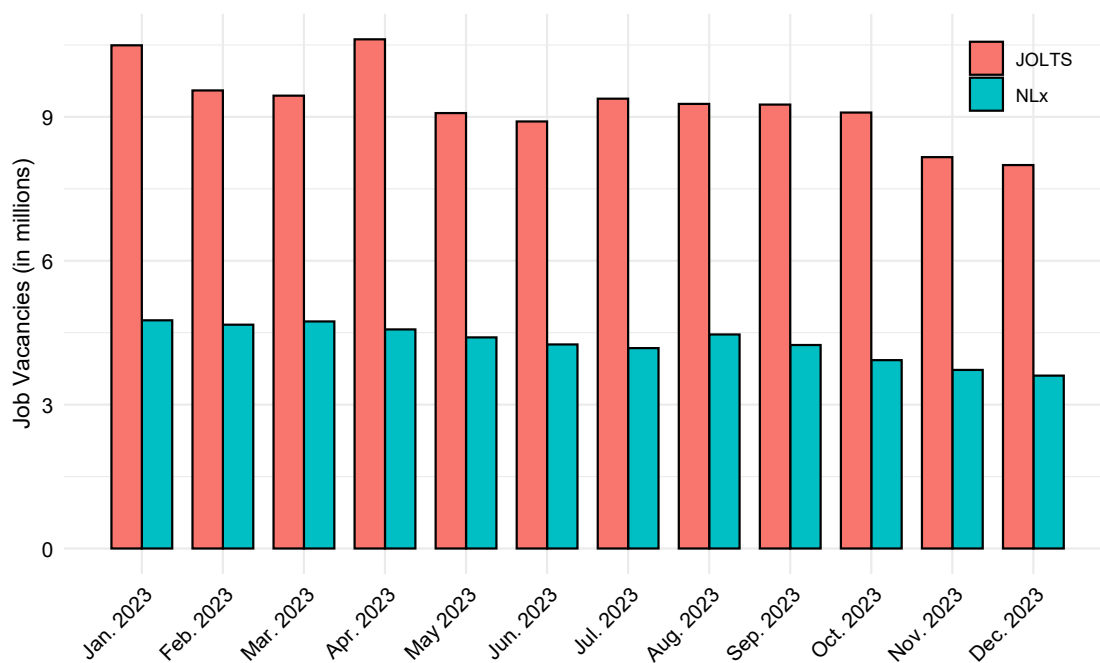
The historical snapshots offer information on all open vacancies on any given day, and the NLx provides pooled monthly versions. To construct the job vacancy distribution, we keep all unique vacancies that open in a given month. We then pool all months to estimate the *distribution* of job vacancies at the state $\times$ occupation level. Vacancies which are open for several months are counted multiple times as in JOLTS. While we could filter these out, including them likely yields a more accurate representation of the vacancies that workers face a given moment within the year. We scale these distributions by the number of job vacancies per unemployed worker from JOLTS and, multiplied with the scaled number of unemployed workers in our ACS sample.

Coverage. The NLx aims to cover all online job postings in the United States, but not all vacancies are posted online. Previous work by [Hershbein and Kahn \(2018\)](#), [Deming and Noray \(2020\)](#), and [Acemoglu et al. \(2022\)](#) has assessed how representative vacancies are, using a commercial dataset by [lightcast.io](#). They found that online vacancies are generally representative in terms of industries and occupations. The main shortcoming – see for example Section 3.A in [Hershbein and Kahn \(2018\)](#) – is that highly skilled occupations tend to be over represented. But we focus particularly on this segment of the market in our application.

To better understand the total coverage we compare the number of vacancies in the NLx and JOLTS. Figure 9 highlights both monthly numbers and the movement over time within our sample year. The NLx data contains on average 4.29 million vacancies per month, representing about 46.3% of JOLTS vacancies. Note that this is the total number, and that coverage in the

subset of highly skilled vacancies will be even higher. The correlation between both datasets over time ($\rho = 0.83$) is also large.

Figure 9: Vacancies in NLx and JOLTS



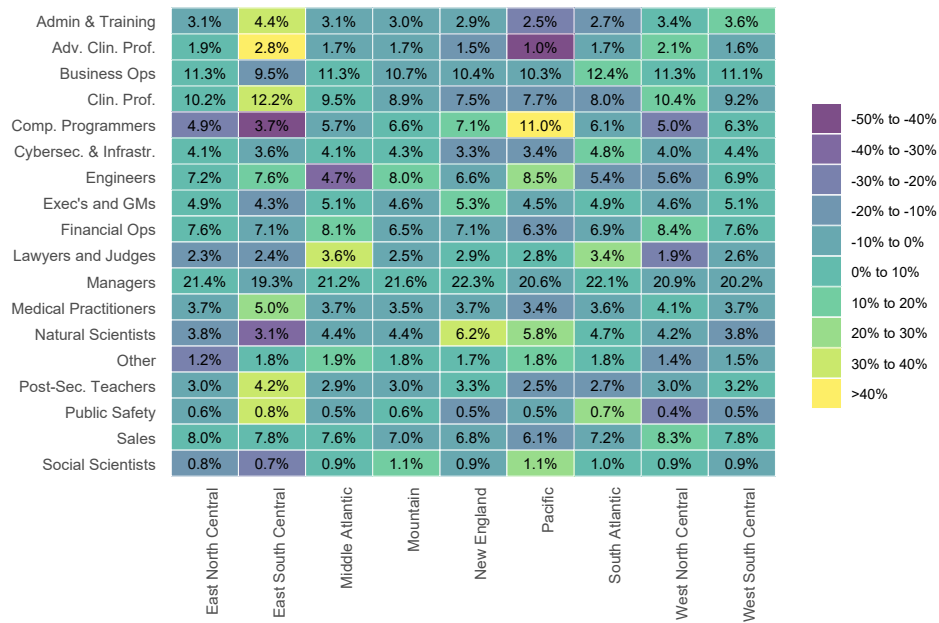
Notes. This figure highlights the monthly number of job vacancies reported by [Job Openings and Labor Turnover Survey](#) (JOLTS) and computed individual postings in the [National Labor Exchange](#) (NLx) Research Hub data.

B.5 Variation in Marginal Distributions Across Markets

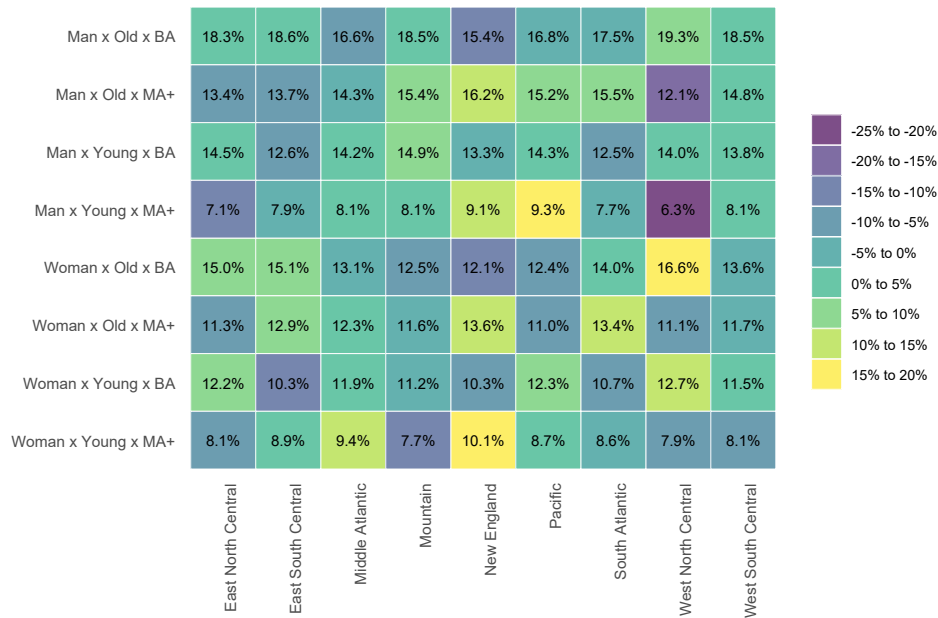
To identify the matching function elasticity, we rely on variation in the marginal distributions of workers and firms across markets. Figure 10 presents these distributions. Panel (a) highlights the differences in worker types. While these are overall relatively similar – as expected – there are some notable differences. For example, Administration and Training occupations represent about 4.4% of the market in the East South Central region, compared to only 2.5% in the Pacific region. The largest difference is for Computer Programmers, who make up 11% in the Pacific region but just 3.7% of the East South Central region. We see the smallest relative variation for Managers, who make up 19.3% of the East South Central region and 22.3% in New England. Panel (b) highlights the comparable, but slightly larger, differences in worker types. For example, young men with at least a Master’s degree account for 9.3% of the Pacific market but just 6.3% in West North Central. Similarly, older men with at least a Master’s degree account for 16.2% of observations in New England, compared just 12.1% in the West North Central region. We conclude that there is sufficient variation in the marginals across these different markets.

Figure 10: Workers and Occupation across Markets

(a) Occupation Types



(b) Workers Types



Notes. This figure shows the number of workers and occupations of each type across the Census Divisions in the United States (2023). The numbers in each cell indicate the fraction of observations within the Census Division that this cell represents. Colors indicate relative deviations from the occupational mean across census divisions. The data is a combination of the [American Community Survey \(ACS\)](#), the [Job Openings and Labor Turnover Survey \(JOLTS\)](#) the [National Labor Exchange \(NLx\)](#) Research Hub job vacancy data.

B.6 Summary Statistics

Table 8 presents further summary statistics about the estimation sample, which is constructed from the 2023 wave of the [American Community Survey](#) (ACS). The sample consists of almost 275,000 individuals and contains slightly more men ($\approx 54\%$) than women, since our focus is on high earning occupations. Because of this sample selection, the unemployment rate is also low ($\approx 2\%$). All individuals in our sample hold a college degree, and 44% have a Master's degree or more, with women being slightly higher educated than men. We selected only individuals ages 25 to 64, and about 58% are 40 years or older, with men being slightly older.

Table 8: Summary Statistics

	Men	Women	Total
Observations			
Employed	148,652	125,892	274,544
Unemployed	2,725	2,519	5,244
Demographics			
Master's Degree +	0.42	0.45	0.44
Age ≥ 40	0.6	0.56	0.58
Occupations			
Business	0.55	0.55	0.55
Health	0.07	0.22	0.14
Other	0.07	0.09	0.08
Science	0.06	0.05	0.06
Tech	0.25	0.08	0.17
Hours			
Full-Time	0.72	0.8	0.76
Long Hours	0.28	0.2	0.24
Wages			
Avg. Month. Wage (in k\$)	12.99	9.59	11.43
Std. Dev. Month. Wage (in k\$)	10.78	7.63	9.62
Avg. Hr. Wage (in \$)	68.12	51.99	60.72
Std. Dev. Hr. Wage (in \$)	53.25	38.23	47.64

Notes. Summary statistics from the latest [American Community Survey](#) (ACS) wave 2023. The selected sample consists of higher educated workers in high earning degrees as introduced in [Goldin \(2014\)](#) and discussed in section 3.

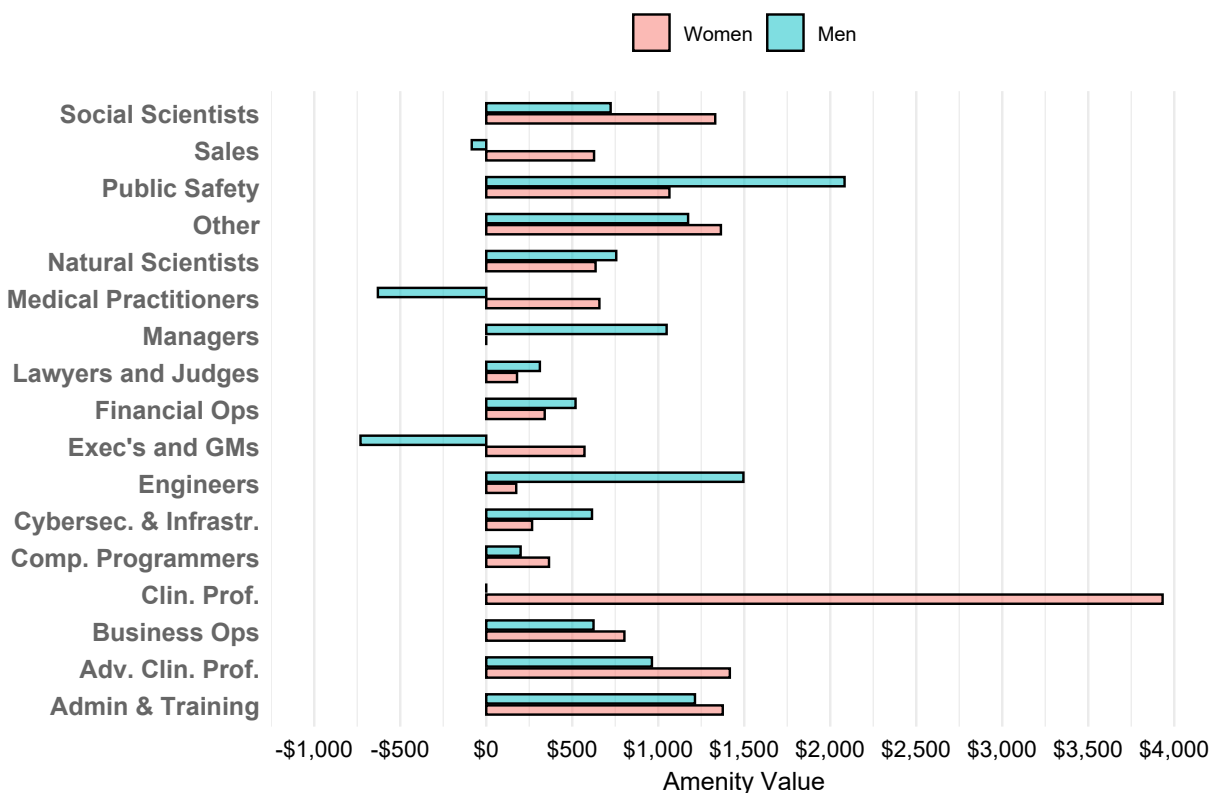
The occupations of men and women are relatively similar, about 55% works in Business, 14% in Health, 17% in Tech, 6% in Science, and 8% in Other occupations. The main difference is that women work more in Health (22% compared to 7%) and men work more in Tech (25% compared

to 8%). We also find that men (28%) are more likely than women (20%) to work long hours. The raw gender gap in monthly earnings is just over 25%, with men earning about 13.000 dollars per month compared to women's 9.500 dollars before taxes. There is also more variance in men's earnings than in women's. The gender gap in hourly wages is just below 25%, with men earning 68 dollars compared to women's 52 dollars, again with higher variance among men.

C Results

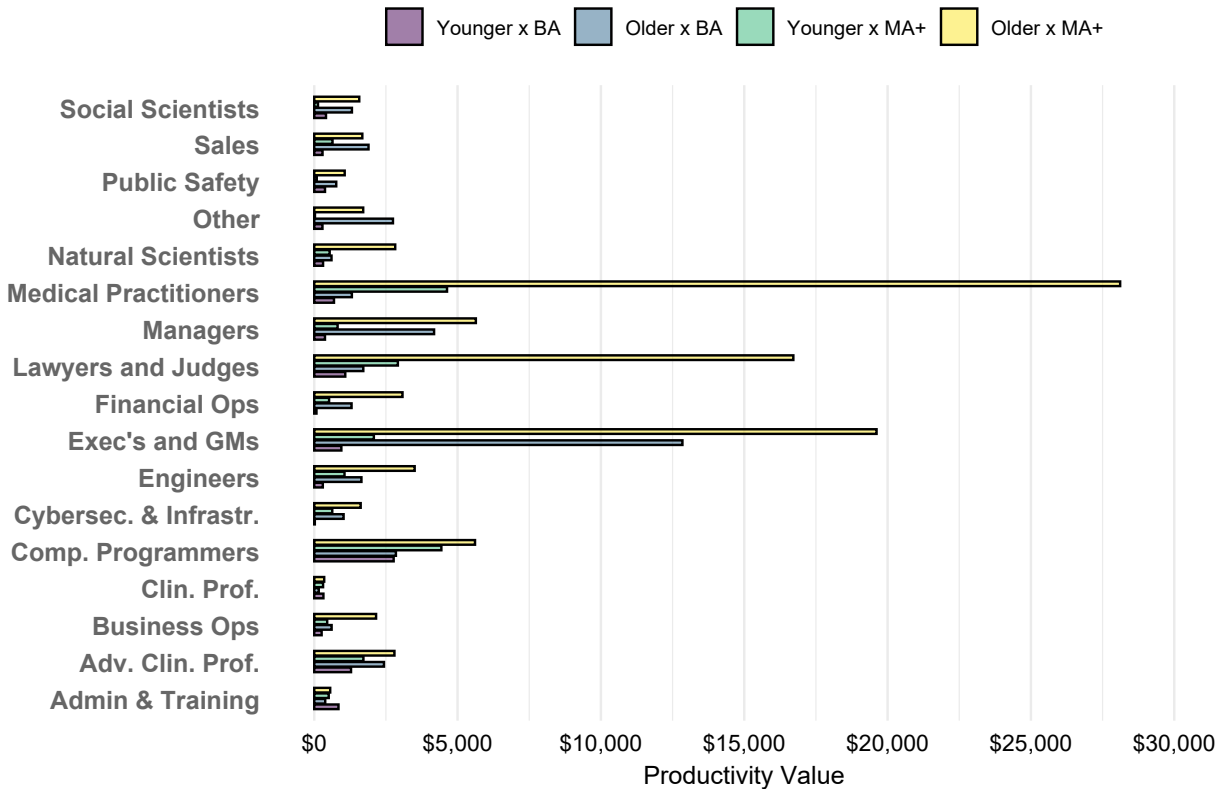
We now present further parameters from the structural model that were omitted from the main text. We first show additional worker-side amenity parameters, then the firm-side productivity values, and finally present the other remaining estimates.

Figure 11: Amenity Value by Gender across Occupations



Notes. This figure shows the estimated amenity value across occupations. For women, the amenities of Managers are normalized to zero. For men the value of Clinical Professionals is normalized to zero. All values are expressed in dollars of monthly earnings.

Figure 12: Productivity Values by Occupations across Workers



Notes. This figure shows the estimated productivity premiums across occupations. The value of Post-Secondary Teachers is normalized to zero. All values are expressed in dollars of monthly earnings.

C.1 Long Hour Premium Fit

We now look at how the model's long hour premiums fit those in data. Figure 13 highlights the model fit. Generally, the model replicates the long-hour wage premiums well, but the fit differs between occupations. The models fits the data particularly well in Business and Tech occupations, where wages are generally set in a competitive manner. The fit is worse in occupations where there is more intervention. The wage premiums in Health occupations are generally over-estimated, except for Medical Practitioners, where model fits well. We also over-estimate the premiums in Teaching and Science occupations. While there are some differences between the data and model, overall we replicate the main patterns.

Figure 13: Model vs. Data Premium

